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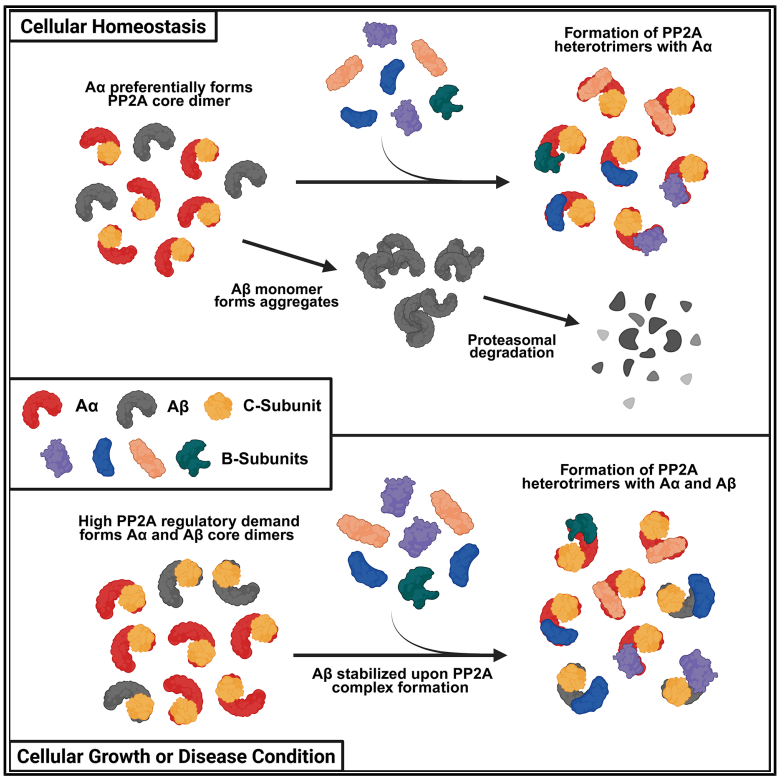
Regulatory mechanisms of PP2A complex assembly driven by physicochemical differences in A-subunit isoforms

Graphical abstract



Highlights

- Cryo-EM structures reveal different conformations of PP2A Aα and Aβ isoforms
- Aβ N-terminus disrupts interactions with B56γ C-terminal helix
- Monomeric Aβ shows lower thermal stability and forms aggregates
- Holoenzyme assembly prevents Aβ aggregation and enhances stability

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In brief

Day et al. demonstrate that PP2A scaffold isoforms Aα and Aβ have different physicochemical properties. Cryo-EM analysis reveals Aβ's more concave structure with disrupted B-subunit interactions. Unlike Aα, monomeric Aβ aggregates readily, though complex formation stabilizes both. Results suggest Aβ serves as a PP2A activity reservoir during periods of high regulatory demand.

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Regulatory mechanisms of PP2A complex assembly driven by physicochemical differences in A-subunit isoforms

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SUMMARY

Protein phosphatase 2A (PP2A) is crucial for regulating cellular pathways, with its holoenzyme assembly affecting enzyme function and substrate selection. The PP2A holoenzyme comprises scaffold A-, regulatory B-, and catalytic C-subunits, each with various isoforms. Here, we examine structural and biochemical characteristics of the A-subunit isoforms (Aα and Aβ) and identify different biophysical properties that may promote distinct PP2A functions. Our molecular dynamics simulations and cryo-EM analyses define structural differences in the isoforms that reside primarily at the N-terminus of the A-subunit where it interfaces with regulatory B-subunits. Kinetic analyses show Aβ has a lower binding affinity in complexes with B56 subunits and exhibits unique aggregative properties as a monomeric protein. These findings suggest that the different physicochemical properties between A-subunit isoforms are key to PP2A holoenzyme assembly and function. We predict that the Aβ serves as a reservoir, ensuring that serine-threonine phosphatase activity is maintained during high regulatory demand.

INTRODUCTION

Protein phosphatase 2A (PP2A) is a family of enzymes that, along with protein phosphatase 1, accounts for approximately 90% of cellular serine-threonine phosphatase activity.[1–3] The crucial roles of PP2A in regulating cellular progression, growth, death, and survival are implicated by consequences in its dysregulation in numerous human diseases, including heart disease, neurodegenerative disorders, and various types of cancer.[4–8] The versatility of PP2A in substrate selection stems from its complex structure. The binding of one of over 40 regulatory B subunits to the AC core, composed of a scaffold A subunit and a catalytic C subunit, influences cellular localization of the holoenzyme and contributes to substrate selectivity.[9] The regulatory B subunits are divided into four structurally distinct families (B55, B56, PR70/72, and STRN) that share little to no sequence similarity among them.[9] Within the family of B subunit isoforms, particularly the B55 and B56 families, protein sequences are highly similar, with the primary differences concentrated at the N- and C- terminal regions.[10] The isoforms within each class also differ in their expression patterns across tissues, as well as in the substrates and pathways they target, allowing for highly localized, regulated, and specific PP2A activity.[11] PP2A's functional diversity is further expanded by the presence of two isoforms of both the A and C subunits (Aα/Aβ and C/Cβ), although their distinct roles within PP2A complex assembly are not yet fully understood. The two C subunit isoforms share 97% sequence identity and are suggested to have largely overlapping functions, even though the Cα isoform is ∼10-fold more abundant and its knockout causes delayed embryonic lethality in mice.[12,13] While reports of cancer-associated mutations are scarce for either isoform, loss of Cβ is observed in myelodysplastic syndrome (MDS) due to genetic deletion.[14] In contrast, both A-subunit isoforms have been identified as tumor suppressors and are frequently mutated in cancer and targeted by viral oncoproteins.[15–19]

Despite sharing 87% sequence identity and possessing well-conserved B and C subunit binding regions,[20] the Aα and Aβ isoforms are differentially bound by viral oncoproteins. The polyoma virus middle T antigen binds PP2A AC heterodimers comprised of either Aα or Aβ subunits, whereas the small T antigen selectively binds to PP2A containing the Aα isoform.[21–23] The primary sequence difference of A subunit isoforms includes a 12 amino acid extension at the N-terminus of Aβ. This subtle variation may contribute to differences in substrate specificity, as exemplified by the RalA GTPase, which is uniquely dephosphorylated by PP2A complexes containing the Aβ isoform.[24] Expression patterns of Aα and Aβ also differ significantly. Whereas Aα exhibits uniformly high expression across tissues, Aβ expression

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varies between tissue types and is generally lower, except in the testis.[22] Moreover, Aα expression remains consistent in tumor cell lines, whereas Aβ expression fluctuates dramatically.[22] The mechanisms of Aα and Aβ inactivation in cancer also differ. Complete loss of Aβ is necessary for transformation, whereas haploinsufficiency of Aα is sufficient.[17,25] This finding is supported by sequencing data showing that homozygous deletions of Aα are rare, while they are relatively common for Aβ due to its location within the frequently deleted 11q23 chromosomal locus.[26–28] Notably, Aα overexpression is unable to compensate for Aβ loss in cellular transformation.[24] On the other hand, knockout of Aα is not only growth suppressive but also results in an increase in Aβ expression and stability.[25] Subsequent knockout of specific PP2A regulatory subunits (e.g., B56γ) in the Aα knockout cell lines results in restoration of growth.[25] Finally, in the regulation of developmental processes, there is evidence for different roles between Aα and Aβ holoenzymes in which a shift from Aβ to Aα expression takes place during Xenopus development.[29] These findings suggest both overlapping and unique functions between Aα and Aβ-containing PP2A holoenzyme complexes.

The general structure of the PP2A A-subunit supports its role as a flexible scaffold protein. It is composed of 15 HEAT repeats that facilitate the tremendous structural adaptability of the A subunit.[30,31] This flexibility is crucial to accommodate the binding of the structurally distinct B subunits and facilitate their interaction with the C subunit to form the many possible PP2A holoenzyme complexes and their substrate-specific binding interfaces.[32] Due to the importance of the A subunit in PP2A holoenzyme assembly, many PP2A-associated diseases result from mutations localized to the A subunit.[33–35] Recurrent mutations, including P179R and R183W, localize to the N-terminal portion of the A subunit, and result in the inhibition of binding of select B subunits or PP2A regulatory proteins.[35] However, these mutants are not always conserved in the two isoforms, indicating specific and non-compensatory roles for Aα vs. Aβ in PP2A function.

Here, we investigated the structural and biochemical properties of the PP2A Aβ isoform, which had been largely unexplored compared to the more studied Aα isoform. Molecular dynamics simulations revealed ranging motions of the extended Aβ N-terminus. The localization of this N-terminal extension is predicted to alter binding interactions between A subunit isoforms and regulatory B subunits. Our cryo-EM map of the PP2A(Aα)-B56γ complex includes features to accommodate intersubunit contacts formed by the B56γ C-terminal helix that extends toward the N-terminus of Aα in the assembled heterotrimeric complex. The absence of this feature in PP2A(Aβ)-B56γ suggests that the Aβ N-terminal twelve-residue extension disrupts this intersubunit interaction. Additionally, in comparison to Aα, the Aβ isoform adopts a more concave conformation in the structure of the assembled heterotrimer, and the cryo-EM map has stronger features to suggest higher occupancy of the C-tail of the C subunit at the A-B56γ interface. Comprehensive binding kinetic analyses demonstrate weaker affinity of the Aβ-containing core dimer to B56α and B56γ isoforms. Finally, we identified a unique propensity for aggregation and weaker thermal stability of the Aβ monomer. Together, these findings reveal a mechanism by which the innate physicochemical properties of the Aβ isoform dynamically self-regulate protein stability and demonstrate a structure-function relationship between A isoform features and complex assembly.

RESULTS

The unique N-terminus of the Aβ isoform is flexible and dynamic

The conformational flexibility of the A subunit allows it to adopt many conformations within the assembled heterotrimer formed with structurally diverse B subunits (Figure S1).[30] The rearrangement is primarily due to changes in a hinge-like region formed at HEAT repeats 11–12 that allows the N-terminal half of the A subunit to adopt different conformations relative to the C-terminal HEAT repeats.[32] Missense mutations in the A subunit of PP2A can restrict protein dynamic motion in various types of cancer, which consequently modifies the binding interactions with regulatory subunits and changes the composition of PP2A holoenzymes within the cell. One example is the recurrent P179R mutation of Aα that has been identified in patients with endometrial cancer.[34] In addition to a unique 12 amino acid extension at the N-terminus of Aβ, the sequences of the Aα and Aβ isoforms include 80 single residue differences, which reside primarily within the helices of HEAT repeats 2–9 and 12–15 (Figure S2). These sequence differences could have an impact on the dynamic motion of each isoform, which could affect the binding interactions formed with other PP2A subunits within an assembled holoenzyme. Thus, to investigate the different structural outcomes resulting from the sequence variations between Aα and Aβ, we conducted molecular dynamics (MD) simulations for each A subunit isoform.

MD simulations were initiated using Aα and Aβ models determined by AlphaFold3 *de novo* prediction. Following a 100 ns simulation, the N- terminus of Aβ exhibits larger ranges of motion throughout the trajectory that is more confined for the Aα isoform (Figures 1A and S3). Dynamic motions predicted from the MD simulations are represented in changes in the RMSD plot of each A subunit isoform relative to the starting structure. While Aα maintains relatively stable RMSD values of around 10-15 Å, Aβ exhibits more pronounced dynamic behavior, gradually increasing from ~10 Å to ~20-25 Å, with notable fluctuations occurring between 20 and 75 ns, followed by a further increase of 25 Å by the end of the simulation (Figure 1B). Particularly, the Aα N-terminus exhibits minimal and isolated motion, whereas the Aβ N-terminus exhibits dynamic motion spanning a larger conformational space (Figures 1B and 1C; Videos S1 and S2). When omitting the N-terminal residues, the RMSD values are similar for both A subunit isoforms. Cumulatively, these data predict that the Aβ N-terminus is dynamic and uniquely occupies a large spatial landscape surrounding the first HEAT repeat of the isoform. Importantly, this region of the A subunit is involved in intersubunit interactions with B subunits in the assembled heterotrimer. Specifically, structures of PP2A holoenzymes determined by X-ray crystallography identify a C-terminal α-helix of the B56γ isoform that extends to form intermolecular contacts with the N-terminus of the Aα subunit.[36,37] Thus, it is expected that these interactions would be altered by the presence of the N-terminal peptide extension of the Aβ isoform (Figure 1D).

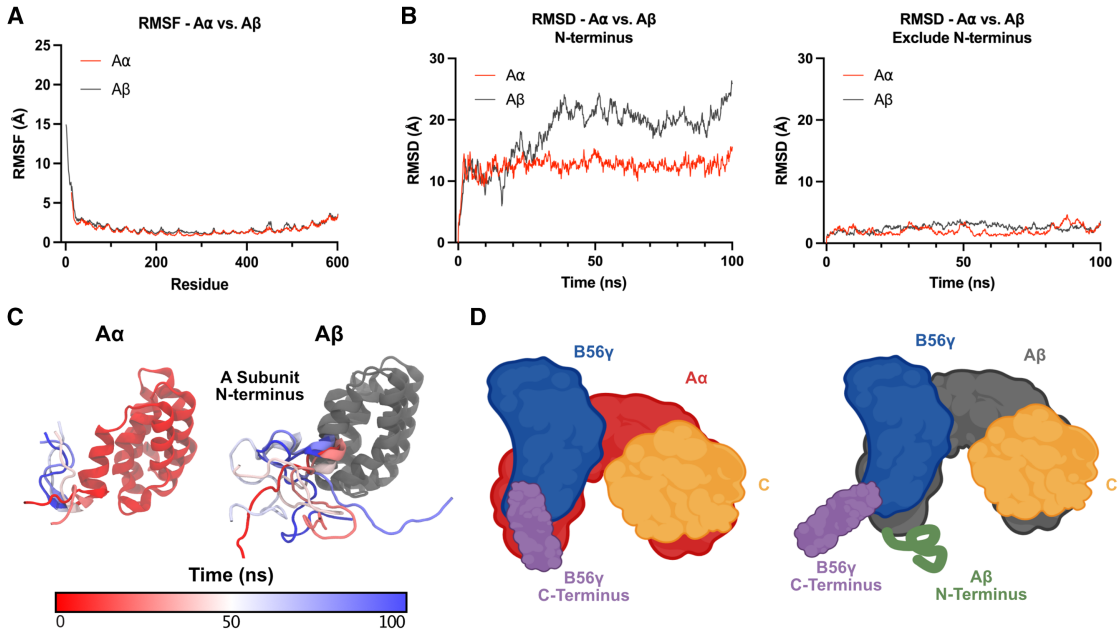

Figure 1. All-atom molecular dynamics simulations of PP2A A subunit isoforms

(A) Root-mean-square fluctuation (RMSF) of the Aα and Aβ subunits over the course of a 100 ns simulation plotted per residue.

(B) RMSD of the N-terminus (residues 1–13 for Aα and 1–25 for Aβ) and entire protein excluding the N-terminus (residues 14–589 for Aα and 26–601 for Aβ) throughout the 100 ns simulation.

(C) Motion of the Aα and Aβ subunit N-terminus (residues 1–13 for Aα and 1–25 for Aβ) throughout a 100ns MD simulation and overlaid at 20ns time increments. Color corresponds to the time point in the simulation from red (0 ns) to blue (100 ns).

(D) Schematic representation of the B56γ C-terminus binding region at the A subunit N-terminus. This interaction is predicted to be interrupted by the extended Aβ N-terminus.

See also Videos S1 and S2; Figures S1–S3.

Structural comparison of Aα and Aβ holoenzymes identifies differences in intersubunit interactions with B56γ

To delineate the molecular interactions in PP2A comprised of Aα or Aβ isoforms, we used cryo-electron microscopy (cryo-EM) and single particle analysis to solve the structure of each isoform assembled as a holoenzyme with the B56γ regulatory subunit (Figure 2). The complexes were assembled and isolated using individually expressed and purified subunits, followed by size-exclusion chromatography (SEC) (Figures S4A and S4B). Approximately 15–20% of the C-subunit included a post-translational carboxymethylation of the terminal leucine residue, as determined by immunoblotting (Figure S4C). Reconstituted PP2A heterotrimers assembled from Aα or Aβ isoforms exhibited similar enzymatic activity when using DiFMUP as a substrate (Figure S4D). The cryo-EM maps were resolved to an average 3.2 Å (PP2A:Aβ-B56γ) and 3.4 Å (PP2A:Aα-B56γ) resolution and were used to build models of the heterotrimeric complexes (Figures S5 and S6). At nominal resolution, the B56γ C-terminal helix interaction with the A subunit was not visualized in either structure. Applying a 6 Å low-pass filter, however, resulted in enhancement of the visibility of this feature in the cryo-EM map for the Aα, but not the Aβ, PP2A complex (Figure 2B). Similarly, 3D classification revealed that the C-terminal helix of B56γ was distinguishable in most sub-classes for the complexes assembled from the Aα isoform, whereas it was missing in all subsets of averaged particles assembled with the Aβ isoform (Figure S7). This observation supports the notion that the additional 12 amino acids at the N-terminus of Aβ disturbs interactions with the B56γ C-terminal helix formed in the context of the assembled Aα-B56γ PP2A complex.

Aligning the A subunits from the two heterotrimer structural models to the C subunit binding region indicates a more concave conformation of Aβ relative to Aα in the assembled heterotrimer (Figure 3A). The more compact conformation of Aβ correlates with an end-to-end distance (as determined by the centers of mass of HEAT repeats 1 and 15) of 59.6 Å versus 62.7 Å for the Aα isoform. Center of mass calculations for individual HEAT repeats indicate that the largest differences in Aα versus Aβ structure lie near the central region of the protein, comprising HEAT repeats 3–6, which are shifted by ∼2-3Å when comparing the two isoforms (Figure 3B). These differences are correlated with features involving a salt bridge that forms between HEAT repeats 3 and 5 in the two structures. Backbone-to-backbone distances of the structural model for this salt bridge (R125 and E203 in Aβ and R113 with E191 in Aα) is ∼1 Å closer in the Aβ structure and is supported by more contiguous features in the cryo-EM map for side-chain accommodations of the residues forming the salt bridge (Figure 3C).

Interestingly, HEAT repeats 3–6, which exhibit the largest differences in overall displacement in structural comparisons of the Aα versus Aβ isoform, are involved in B subunit interactions in the assembled heterotrimer. Despite this difference, most of the interactions that form the PP2A A-B56γ interface are largely

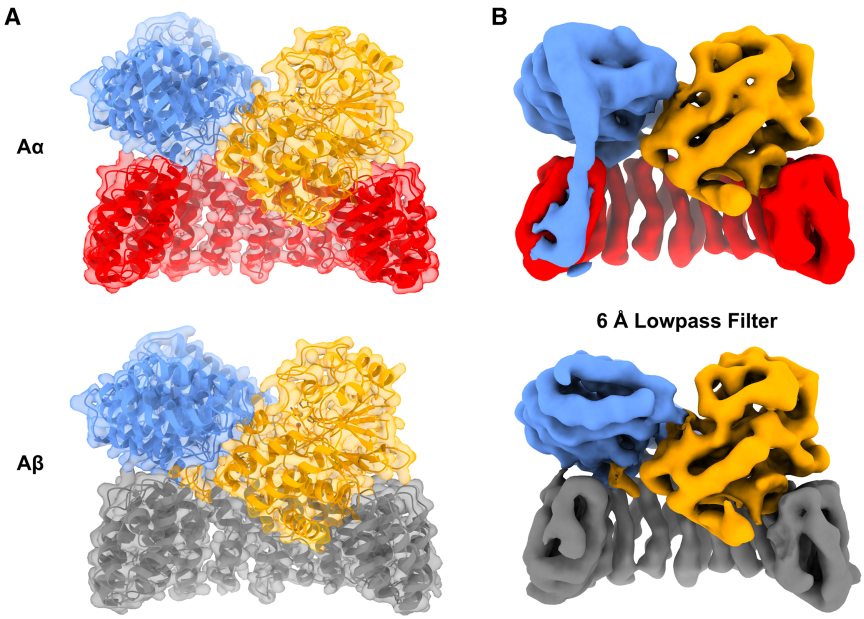

Figure 2. Structures of PP2A heterotrimers assembled from different A subunit isoforms identify a major difference in intersubunit interactions

(A) Sharpened cryo-EM maps using EMReady for both the Aα-B56γ-C and Aβ-B56γ-C PP2A holoenzyme complexes (level = 3.7) with refined models fit within the maps.

(B) Low pass filter at 6 Å of the Aα-B56γ-C and Aβ-B56γ-C PP2A holoenzyme complex raw cryo-EM maps reveals features to accommodate the B56γ C-terminal helix only in the PP2A complex assembled with Aα. In all panels, Aα is colored red, Aβ is colored gray, B56γ is colored blue, and C is colored orange. See also Figures S4–S6; Table S1.

conserved in both isoforms (Figure 3D). Specifically, this interface involves intra-repeat loops of A subunit HEAT repeats 2–7 and includes E100/E112, E101/E113, R183/R195, and W257/W269 using Aα/Aβ identification, each of which interacts with the same residues in several helices of the B56γ subunit. Similarly, residues composing the A-C interface are conserved in Aα and Aβ and include molecular contacts from intra-repeat loops and inner helices of HEAT repeats 11–15 (Figure 3E). The A subunit residues that contribute to the A-C interface include K416/K428, R418/R430, Y456/Y468, Y495/Y507, R498/R510, D572/D584, D574/D586 using Aα/Aβ identification.

Trimeric interface differs in A subunit isoform complexes

The binding of certain regulatory B subunits is dependent on a unique leucine carboxymethylation event on the conserved, terminal leucine residue of the catalytic C subunit.[38,39] Structures of PP2A-B56 holoenzymes reveal that the last ten residues of the C subunit form a tail (C-tail) that can bind in a groove formed at the A-B56 interface, interacting with both subunits to stabilize the assembled complex.[10,40] When positioned in this groove, methylation of the terminal leucine residue of the C-tail neutralizes a charge repulsion associated with the non-modified, carboxyl leucine residue. Thus, C-tail methylation of PP2A is a key regulator in the selection of specific B subunit holoenzyme complexes. While binding of the B56γ subunit does not depend on this methylation event, the extended C-tail is required for holoenzyme assembly.[39] Comparison of our PP2A:Aα-B56γ and PP2A:Aβ-B56γ holoenzyme structures reveals that the entirety of the C-tail can be traced in the Aβ cryo-EM map, whereas analogous features to accommodate C-tail modeling in the Aα map are fragmented, even following 3D classification procedures (Figure 4). After 3D classification, nine of the ten classes within the Aβ dataset and only three of the ten classes within the Aα dataset displayed features to accommodate the C-tail (Figure S7). Interestingly, the classes in the Aα dataset that displayed C-tail density similarly displayed density for the B56γ C-terminal helix. This finding identifies a conformational correlation between the placement of the B56γ C-terminal helix and the C subunit C-tail, either of which could conceivably contribute to the selection of B subunits and/or the stability of the assembled heterotrimer.

Aβ displays weaker binding affinities to B56 isoforms relative to Aα

The binding affinities of B subunits to the PP2A AC core dimer contribute to regulating the population of PP2A holoenzymes formed, which, in turn, dictates substrate selection and phosphatase activity of PP2A in the cell. Previous cellular studies demonstrated that, under overexpression conditions, B56α binding is weaker for Aβ than for Aα isoforms, suggesting differential binding preferences between scaffold isoforms.[22] A subunit binding to B56γ also exhibits a distinct behavior, where depletion of Aα enhances B56γ interaction with Aβ.[25] Notably, this enhanced B56γ-Aβ interaction is crucial for Aβ-specific growth suppressive activity in the context of Aα loss, highlighting the functional significance of this regulatory interaction.[25] Quantitative binding of these two B56 isoforms with Aα versus Aβ scaffolds would help to understand differences and contributions to PP2A assembly and cellular growth regulation. Thus, we performed quantitative binding studies using biolayer interferometry (BLI) to define the binding of AβC or AαC core heterodimers with B56α and B56γ regulatory subunits. B56 subunits were immobilized on the BLI tip, and the AβC or AαC core dimers were used as the analyte. Overall, AβC was measured to have weaker binding affinities relative to AαC, for both B56 subunits tested (Figures 5 and S8A). After fitting the experimental data, the equilibrium dissociation constant (K_D) of the B56γ subunit binding to AαC and AβC was determined to be 9.2 nM and 14.0 nM, respectively. The K_D for B56α binding to AαC and AβC was determined to be 25.7 nM and 119.9 nM, respectively.

To define the precise contribution of the Aβ N-terminus to B56 subunit binding affinity, we constructed and purified two modified A subunit proteins. In one construct, the Aβ N-terminus was removed to generate an Aβ N-truncation protein (referred to as AβNT). In the second construct, the unique 12 amino acid

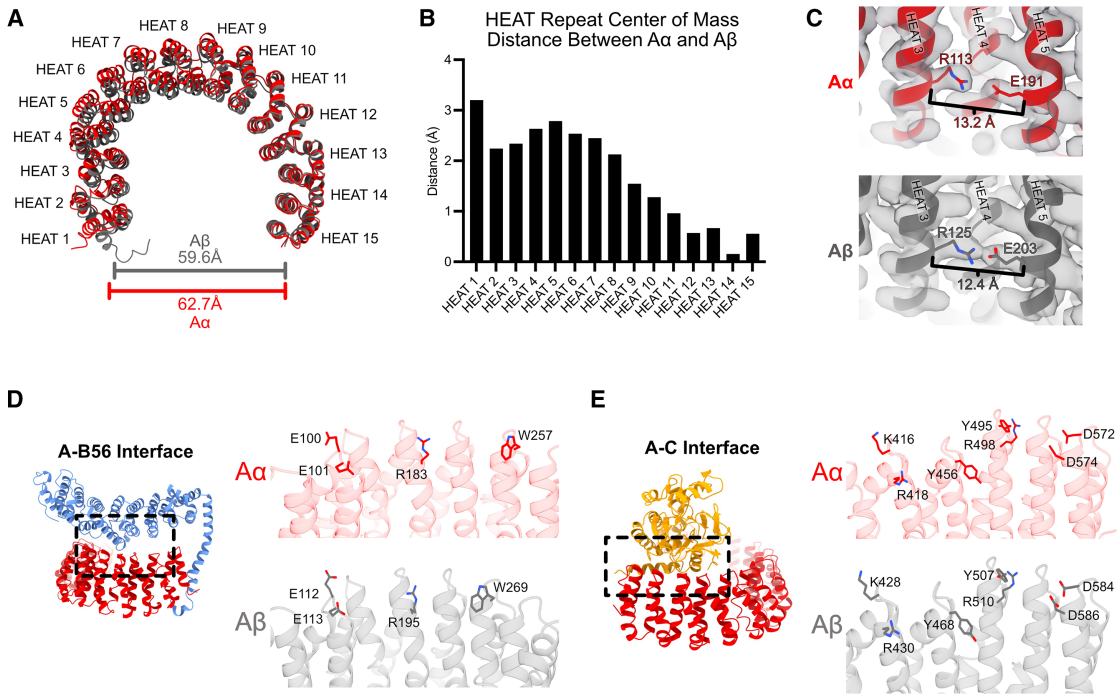

Figure 3. Structural comparison and intersubunit interface of PP2A complexes assembled with B56γ and Aα or Aβ isoforms

(A) Superimposed A subunit isoform models of PP2A:Aα-B56γ and PP2A:Aβ-B56γ aligned to the C-subunit binding region (HEAT 12–15). Measured distance between the center of mass of HEAT repeat 1 and 15 for both Aα and Aβ models is shown.

(B) Difference in distances between the center of mass of each HEAT repeat measured between Aα and Aβ.

(C) Comparison of inter-HEAT repeat interaction at the salt bridge between repeats 3–5 in Aα and Aβ. Backbone-backbone distance measured between HEAT repeats 3 and 5 at the salt bridge forming residues (R113 with E191 in Aα and R125 with E203 in Aβ) is labeled.

(D) Residues composing the B subunit binding interface between A subunits and B56γ.

(E) Residues composing the C-subunit binding interface between A subunits and Cα. In all panels, Aα is represented in red and Aβ in gray.

See also Figure S2.

Aβ N-terminal extension was fused with Aα to engineer a chimera Aα protein (referred to as AαChimera). Each of the modified A subunit proteins were incubated with the C subunit and the assembled heterodimers were subsequently purified by SEC (Figure S8B). We again performed BLI experiments to define the binding of B56α and B56γ subunits to the AαChimera-C and AβNT-C core dimers. In comparison to wild type (WT) Aα-C, the AαChimera-C had a slightly weaker binding affinity

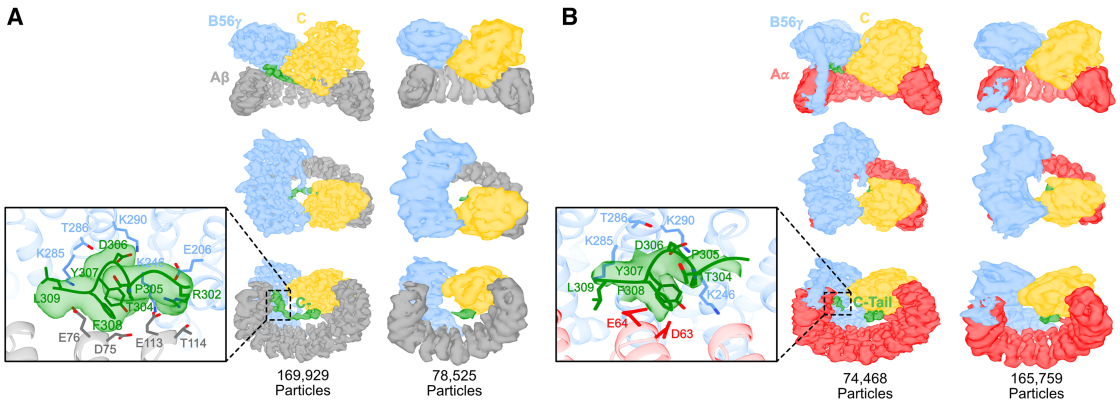

Figure 4. Occupancy of the C-terminal tail differs in the cryo-EM maps of PP2A:Aα-B56γ and PP2A:Aβ-B56γ PP2A complexes

(A and B) Refined maps resulting from 3D classification of particles from the final refined cryo-EM maps for (A) PP2A:Aα-B56γ and (B) PP2A:Aβ-B56γ datasets. Features in either map attributed to the C-tail are colored green. In the zoom-in views, residues surrounding the modeled C-subunit tail are labeled and color coded for each subunit. See also Figure S7.

A

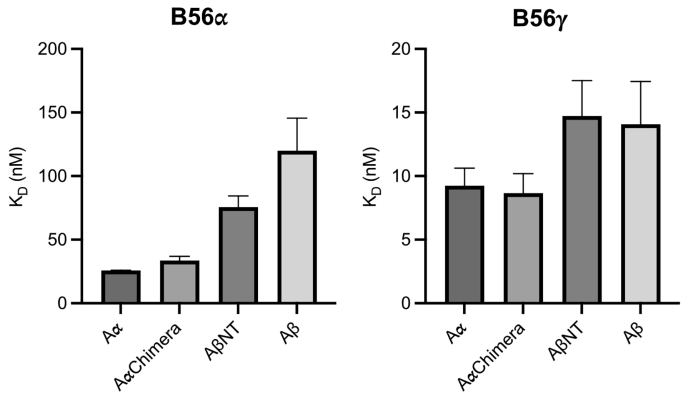

Figure 5. Kinetics of B56α and B56γ regulatory subunits binding to AC core dimers assembled with Aα or Aβ subunit Isoforms

(A) Binding affinities of B56α and B56γ regulatory subunits for PP2A dimers assembled with Aα or Aβ isoforms. Measured AαC and AβC binding to BLI surface immobilized with B56α or B56γ by amine crosslinking (n = 2, the error bars show mean ± SD).

(B) Summary of the kinetic parameters determined from fitting the association and dissociation curves from BLI analysis of Aα, AαChimera, Aβ and AβNT PP2A core dimers binding to B56α and B56γ (n = 2, all reported values are averages, errors for K_D values are SD). Individual experiment curve-fitting statistics are shown in Figure S8.

B

	B56α			B56γ		
	K_D (nM)	k_a ($M^{-1}s^{-1}$)	k_d (s^{-1})	K_D (nM)	k_a ($M^{-1}s^{-1}$)	k_d (s^{-1})
Aα-C	25.7 ± 0.28	2.62 x 10^5	6.73 x 10^{-3}	9.24 ± 1.4	1.78 x 10^5	1.67 x 10^{-3}
AαChimera-C	33.6 ± 3.28	2.22 x 10^5	7.41 x 10^{-3}	8.66 ± 1.54	1.86 x 10^5	1.63 x 10^{-3}
Aβ-C	119.9 ± 25.67	1.13 x 10^5	1.37 x 10^{-2}	14.07 ± 3.36	1.78 x 10^5	2.53 x 10^{-3}
AβNT-C	75.4 ± 8.83	2.03 x 10^5	1.52 x 10^{-2}	14.73 ± 2.79	1.70 x 10^5	2.54 x 10^{-3}

to B56α (34 nM vs. 26 nM) and similar binding affinity to B56γ (8.7 nM vs. 9.2 nM) (Figures 5 and S8A). In comparison to WT Aβ-C, the AβNT-C protein had a tighter binding affinity to B56α (75 nM vs. 120 nM) and similar binding affinity to B56γ (14.7 nM vs. 14.1 nM) (Figure 5). These findings suggest that the unique N-terminus of the Aβ isoform affects the binding affinity of some, but not all, B56 isoforms in PP2A complex assembly.

The PP2A Aβ isoform uniquely forms oligomer aggregates

The genetic removal of Aα, at least in cancer cells, is compensated by a dramatic increase in expression of Aβ, resulting in an increase in Aβ specific holoenzymes and an Aβ half-life that quadruples from 6 to 24-h[25] This finding suggests that Aβ is stabilized through the increased formation of PP2A holoenzyme complexes that occur in the absence of the Aα isoform. Here, we investigated the physicochemical properties of PP2A A isoforms as a monomer and in the context of an AC heterodimer. We had noticed that isolation of recombinant monomeric proteins by ion-exchange chromatography (IEX) resulted in two distinct peaks for Aβ, whereas the Aα isoform eluted as a single peak under identical conditions (Figure 6A). The first Aβ peak (peak 1) eluted at the same conductivity as Aα (~29 mS/cm) and was used for our structural studies described above. However, a second peak (peak 2) eluted at higher conductivity (~35 mS/cm). Thus, to define the differences in the Aβ IEX peak populations, we further evaluated the individual peak fractions using size exclusion chromatography (SEC). The Aβ peak 1 and the single Aα peak from the IEX analysis each elute as a single peak (at ~1.2 mL) on the SEC chromatogram and correspond to the expected mass of the ~65 kDa A subunit. The Aβ peak 2 from the IEX column, however, exhibited an SEC chromatogram with two distinct peaks, one at the expected mass of ~65 kDa and a second peak that eluted in the void volume (Figure 6B). The population in the void volume is indicative of protein aggregation, which was supported by the appearance of larger complexes in negative stain electron micrographs of the Aβ IEX peak 2 fraction that were not apparent in the peak 1 IEX fraction (Figure 6C). Taken together, these data indicate that purified Aβ, but not Aα, protein is prone to forming oligomer aggregates that can be separated from the monomeric protein.

We next assessed the ability of Aβ protein isolated from peak 1 or peak 2 of the IEX chromatogram to bind C-subunit to form an AC core dimer. To do so, each peak fraction of Aβ protein isolated on an IEX column was individually incubated with excess C-subunit and subsequently analyzed by SEC chromatography (Figure 6B). The peak 1 IEX fraction showed a leftward shift of the primary Aβ monomer peak previously identified in conjunction with the appearance of the excess C-subunit monomer peak. This observation supports the formation of the higher molecular weight AβC core dimer. In the case of the peak 2 fraction from the IEX chromatogram, the primary Aβ monomer peak similarly shifted to the left, whereas the void volume peak remained unchanged in the SEC profile after incubation with free C-subunit (Figure 6B). These data further indicate that recombinant Aβ protein is prone to aggregation and, once aggregated, it is unable to bind the C-subunit.

To further analyze the aggregation properties of Aβ compared to Aα in a time-dependent manner, Aβ monomer was purified by IEX as peak 1 and Aα again purified as a single peak. Purified Aα and Aβ monomer proteins were separately incubated at 37°C for 6 h, with samples taken in one-hour increments and subsequently analyzed by SEC. The results identify a decrease in the Aβ monomer peak over time in conjunction with an increase in the void volume peak that is indicative of protein aggregation (Figure 7A). The area under the curve (AUC) of the monomer peak was calculated for each sample and plotted over time (Figure 7B). A clear difference emerges between Aα and Aβ by the three-hour time point in which ~70% of Aβ protein exists in its monomeric form as compared to 96% for Aα. After six hours, the monomer peak of Aβ decreases further to ~50% remaining compared to ~95% of Aα. To dissect whether the aggregation

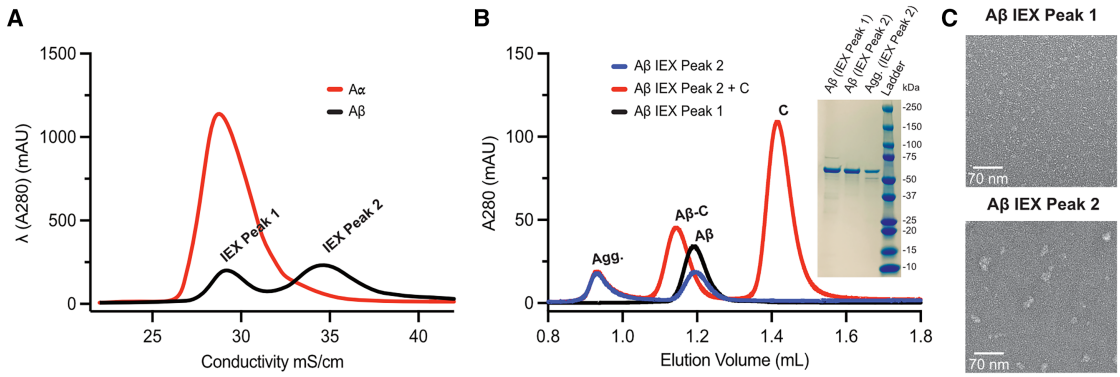

Figure 6. Unique aggregation property of the Aβ isoform

(A) Ion exchange (IEX) chromatography peaks from the purification of Aα and Aβ. Two peaks were observed for Aβ, while only one peak was observed for Aα. One of the two Aβ peaks eluted at the same column volume as Aα (~29 mS/cm), while the other eluted later (~35 mS/cm).

(B) Samples from the Aβ IEX peaks analyzed using size exclusion chromatography (SEC). The first IEX peak sample produced a single peak by SEC at ~1.2 mL (black), corresponding to the Aβ monomer. The second IEX peak sample produced two peaks by SEC (blue), one in the void volume fraction (~0.8 mL), corresponding to protein aggregate, and one at ~1.2 mL, corresponding to the Aβ monomer. Denaturing SDS PAGE gel of SEC peak fractions reveals a protein at ~65kDa for both peaks. IEX peak 2 was combined with excess PP2A C-subunit *in vitro* and analyzed by SEC (red) to reveal three peaks: one in the void volume (~0.8 mL) corresponding to protein aggregate, another at ~1.15 mL corresponding to the assembled AβC core dimer, and a third at ~1.45 mL corresponding to the unbound excess PP2A C subunit monomer.

(C) Negative stain images of samples analyzed from the two ion exchange peaks observed during Aβ protein purification.

properties of Aβ protein are related to the unique N-terminus extension, we repeated the aggregation time-course experiment with the AβNT and AαChimera proteins (Figures 7B and S9). Analysis by IEX indicated that the AαChimera-C protein elutes as a single peak at the same ionic concentration as WT Aα, whereas the AβNT-C protein elutes as two peaks in the IEX chromatogram, similar to WT Aβ protein (Figure 7C). As these mutant proteins mimic their wild type counterparts, we conclude that Aβ aggregation is regulated by regions of the protein that do not involve its unique N-terminal extension. We next asked whether the aggregation of Aβ is unique to the monomeric protein or whether it is maintained when assembled as an AC core heterodimer. To this end, we performed an aggregation experiment with the AαC and AβC core dimer complexes. Each of the A subunit isoforms was combined with molar excess of the C subunit and the assembled complex was purified by SEC. The complexes were once again incubated at 37°C, with aliquots taken every hour for six hours and subjected to SEC to identify the degree of aggregation over time (Figure 7D). Interestingly, the rate of aggregation was similar for both Aα and Aβ when bound to the C subunit (Figure 7E).

We next performed thermal shift assays to determine whether aggregation of Aβ was associated with protein stability. Thermal stability was measured for Aα, AαChimera, Aβ, and AβNT monomer protein. The melting temperature of Aα and AαChimera was 52.1 ± 0.3°C and 52.3 ± 0.3°C respectively, while Aβ and AβNT was 39.4 ± 0.3°C and 39.8 ± 0.4°C respectively (Figure 7F). These data suggest that the N-terminus extension of Aβ does not contribute to the thermal stability of the protein even though Aβ is significantly less thermally stable than the Aα isoform. To define protein stability with respect to complex assembly, we also measured the thermal stability of both the AαC and AβC core dimer complexes after purification by SEC. The AβC core dimer was significantly less thermally stable than the AαC core dimer, as indicated by a melting temperature of 46.7 ± 0.7°C vs. 56.8 ± 0.3°C, a difference similar to that observed for the Aα and Aβ monomers (Figure 7F). Finally, we measured the thermal stability of the PP2A:Aα-B56γ and PP2A:Aβ-B56γ heterotrimeric complexes. The PP2A:Aβ-B56γ complex exhibited a primary change at 51.2 ± 0.1°C vs. 56.7 ± 0.4°C for the PP2A:Aα-B56γ complex (Figures 7F and S10).

DISCUSSION

Assembly of PP2A involving diverse holoenzyme complexes is facilitated, at least in part, by the physicochemical properties of the PP2A A subunit. Both isoforms have been separately defined as tumor suppressors, and mutations in both isoforms have been identified in human cancers. Interestingly, despite sharing 87% sequence identity, previous work has suggested that Aα and Aβ are not entirely functionally interchangeable, and unique substrates have been reported for each isoform.[21,22,24,25,41,42] Furthermore, the Aβ isoform displays a unique expression pattern that differs from Aα, with overall lower and more variable expression across tissues. Physicochemical and structural studies of PP2A have thus far been limited to PP2A complexes containing the Aα isoform. Here, we reveal structural and physicochemical characteristics of the PP2A A-subunit Aβ isoform that likely contribute to functional differences as compared to the more extensively studied Aα isoform.

A remarkable difference between the isoforms relates to thermal stability and propensity for aggregation of the monomeric proteins. We show that the Aβ monomer aggregates rapidly and is significantly less thermally stable relative to Aα. The presence or absence of the Aβ N-terminus on either isoform did not affect these properties relative to the wild type isoforms. The lower thermal stability of the Aβ monomer suggests the isoform is more readily unfolded, which may explain the propensity of the Aβ monomer to aggregate. Notably, we show that the Aβ aggregates do not form PP2A complexes, suggesting these

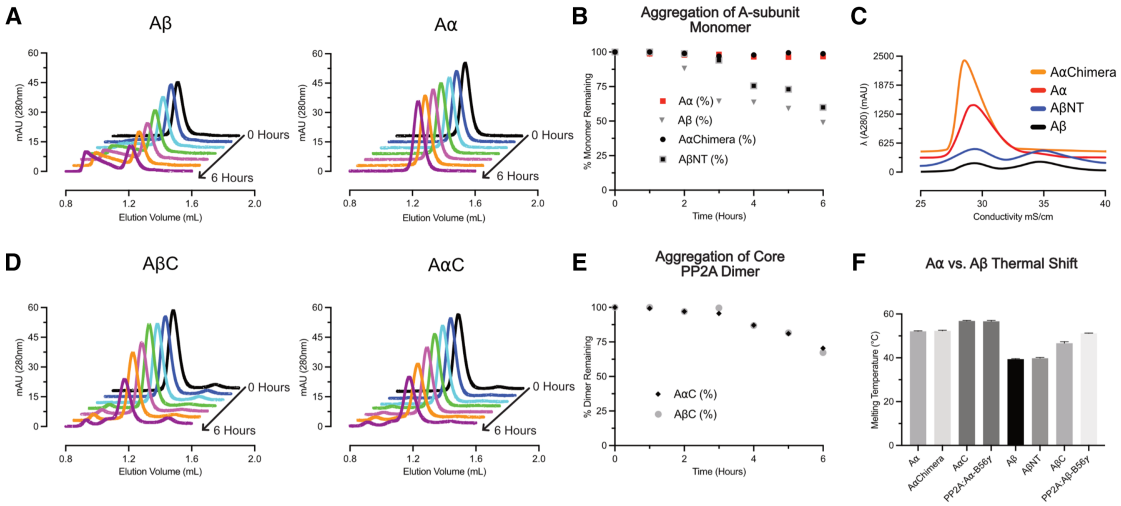

Figure 7. Aggregation and thermal stability differences between Aα and Aβ

(A) Equal concentrations of Aβ, AβNT, Aα, and AαChimera were incubated at 37°C with aliquots taken every hour for 6 h and analyzed by size exclusion chromatography (SEC). The area under the curve (AUC) for Aβ are calculated and plotted over time to reveal a decrease in the Aβ monomer peak with a corresponding increase in the void volume (aggregate) peak.

(B) The AUC for the monomer peak was measured and plotted at each time point relative to timepoint zero giving the value of % monomer remaining for the different constructs.

(C) IEX peaks from the purification of Aα, and Aβ, as well as the AαChimera and AβNT mutants. Two peaks were observed for both Aβ and AβNT, while only one peak is observed for both Aα and AαChimera.

(D) Equal concentrations of the AαC and AβC core dimers were incubated at 37°C and samples were taken every hour for 6 h and analyzed by SEC. The resulting curves for AαC are plotted showing a decrease in the core dimer peak over time with a corresponding increase in the void volume peak.

(E) The AUC for the core dimer peak was measured and plotted at each time point relative to timepoint zero giving the value of % core dimer remaining.

(F) Melting points determined from thermal shift analyses comparing Aα vs. Aβ as a monomer, core AC dimer, and PP2A:B56γ holoenzyme complex. Error bars represent standard error between three replicates of each measurement.

See also Figures S9 and S10.

aggregates may correlate with protein unfolding. Since misfolded protein aggregates are often degraded through the ubiquitin-proteosome system, this could explain a possible innate regulatory mechanism for Aβ isoform expression levels in the cell. In contrast to the monomer protein, the assembly of Aβ into a PP2A complex reduces its aggregation properties and show that the physicochemical properties of the A isoforms are rather similar when assembled as an AC dimer. In this context, a disease state resulting in lower levels of Aα (e.g., due to mutation) or a regulatory event requiring elevated levels of overall or Aβ specific PP2A activity (e.g., during development) would allow Aβ to form more functional PP2A complexes. The increased formation of Aβ PP2A complexes would stabilize the isoform by preventing aggregation, thereby increasing Aβ protein levels and half-life in the cell. In this model, Aβ may function as a reservoir for those instances where additional A subunit is needed for PP2A activity. These observations are consistent with previous findings showing that Aβ protein levels and stability are increased upon Aα knockout in cells in the absence of changes in Aβ mRNA levels.[25] As such, the variable expression levels of Aβ between tissues, in which tissues that show a high level of Aβ could be due to a high demand for Aβ-specific activity, and would result in Aβ stabilization by increased holoenzyme assembly.[22] Aβ is expressed at high levels in the testis, where it is involved in the regulation of spermatogenesis, which requires higher proliferation rates and increased phosphatase activity.[22,41] Together these findings describe a model by which the innate physical characteristics of the Aβ isoform allow for dynamic self-regulation of the Aβ protein level in the cell.

The structural similarities of PP2A heterotrimeric complexes might suggest that Aα and Aβ isoforms are functionally interchangeable. However, previous studies indicate that Aα is unable to substitute for Aβ in cells transformed by Aβ loss and Aβ is implicated in unique regulation of substrates including AKT and RalA and those involved in the spermatogenesis pathway.[21,24] A slightly more concave conformation of Aβ coincides with other differences in the holoenzyme structures, including a higher occupancy of the C-terminal tail of the C subunit within a groove formed at the A-B56γ interface of PP2A. This difference in C-tail occupancy could conceivably contribute to the binding of B subunits for AC dimers composed of the different A subunit isoforms. A higher occupancy of the C-tail at the A-B56 interface in the Aβ structure would be expected to provide added contacts between A and B subunits and, thus, increase the affinity of B subunit binding when compared to PP2A complexes assembled with the Aα isoform. However, our quantitative binding data indicate that PP2A assembled with the Aα isoform exhibit a higher affinity for both B56α and B56γ complexes when compared to those assembled with the Aβ isoform. Our molecular dynamics simulations predict that the extended Aβ N-terminus is dynamic and could interrupt binding interactions with protein-protein contacts involving B56 regulatory subunits and the N-terminus of A subunit isoforms. This notion is supported by our cryo-EM structures of the PP2A:B56γ heterotrimers and, specifically, the complete

absence of features in the cryo-EM map to accommodate the B56γ C-terminal helix only in the Aβ heterotrimer. This difference suggests that the Aβ N-terminus contributes to binding interactions of some regulatory subunits. Thus, the presence of the extended N-terminal peptide of the Aβ subunit might adversely alter the binding of other B subunits, especially for striatins, or those that are part of the integrator complex (INTAC). Structures of these PP2A complexes reveal key interactions with the A subunit N-terminus (Figure S1).[37,43]

Unexpectedly, we show that the binding affinity of B56 subunits are differentially affected by the presence of the N-terminal peptide that is unique to the Aβ isoform. Additionally, the B56γ C-terminal helix interaction with the A subunit N-terminus does not contribute significantly to the binding interaction. In contrast, the B56α binding affinity was affected by the Aβ N-terminus, resulting in a weaker binding affinity when it was added to Aα and a stronger binding affinity when removed from Aβ. Considering that all the B56 class isoforms similarly contain sequences predicted to form an analogous helix at the C-terminus, these observations suggest a differential dependence on the binding interaction of the C-terminal helix of B56 isoforms and the A subunit N-terminus. Since the B56α isoform is methylation-dependent, the binding affinity of B56α is significantly weaker in the presence of a non-methylated C-tail, while B56γ binding is unaffected by C-tail methylation. These observations indicate a weaker binding interaction of the methylation-dependent B subunits with the AC core dimer at baseline, which facilitates this regulatory mechanism of PP2A holoenzyme assembly. Since our experimentation was performed in the context of a partially methylated (15–20%) C subunit, these observations suggest a dependence on the B56γ C-terminal helix binding interaction with the A subunit N-terminus specifically for methylation-dependent B56 subunits that may rely on this additional interaction. Notably, the effect of Aβ excluding the extended N-terminus was more significant than the N-terminus itself in the B56α binding interaction. This finding suggests that differences in Aβ outside of the extended N-terminus result in a weaker binding affinity to B56α. It is possible that the conformational differences identified in the B subunit binding region of the PP2A:Aβ-B56γ complex might have a more significant effect on the binding with B56α, especially considering the additional dependence on methylated C-tail binding at this interface. The extended N-terminus of Aβ may additionally play a role outside of the proposed facilitation or inhibition of binding with specific substrates or B subunits.

Aside from the N-terminus, Aβ differs from Aα at 83 residue positions that exist entirely outside of the intra-HEAT repeat loops that form the binding interfaces with the B and C subunits (Figure S2). The recurrent endometrial cancer mutant, P179R, of the PP2A A subunit Aα isoform dramatically alters the conformational dynamics of the protein and impairs holoenzyme formation with the C and many B subunits.[34] Other recurrent Aα mutants such as R183W and S256F similarly result in significantly impaired holoenzyme formation.[34,35] These mutants highlight the significant impact of subtle changes to the Aα isoform sequence, even though our cryo-EM structures indicate that Aα and Aβ are quite similar in the assembled holoenzyme complex with B56γ. Outside of this assembled state, these residue differences could feasibly result in significant conformational changes between the isoforms that result in altered or impaired binding to specific regulatory B subunits or substrates. Differential binding affinities to both B and C subunits have also been reported for Aβ versus Aα isoforms in a cellular context.[22,25,44] Conceivably, the altered substrate and B subunit binding affinities result from other unique structural characteristics of Aα versus Aβ.

Altogether, our work identifies unique structural and physical characteristics of the PP2A Aβ isoform and defines a role in the regulation of Aβ holoenzyme assembly. We provide insight into the structural basis for the regulation of specific B subunit binding driven by the extended Aβ N-terminus as well as identify unique physicochemical properties of aggregation and stability regulated by PP2A holoenzyme assembly. We further demonstrate conformational differences of Aβ in the PP2A:B56γ structures and provide insight into their role in regulating B subunit binding. Finally, our investigation proposes a novel mechanism by which Aβ protein levels are regulated through a controlled aggregation process with the monomer population being available for instances of specific or high PP2A activity demand.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and agents should be directed to and will be fulfilled by the lead contact, D.J.T. (djt36@case.edu).

Materials availability

Plasmids generated during this study can be requested from the lead contact, Derek Taylor (djt36@case.edu).

Data and code availability

Atomic coordinates and cryo-EM maps reported in this paper have been deposited into the Protein DataBank, PDB: 9MIP (Aα-B56γ-Cα), PDB: 9MF5 (Aβ-B56γ-Cα) and Electron Microscopy DataBank, EMDB-48300 (Aα-B56γ-Cα), EMDB-48224 (Aβ-B56γ-Cα). They are publicly available as of the date of publication. The accession numbers are also stated in the key resources table. This paper does not report original code. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

All authors conceived the study. A.D. and W.H. solved and analyzed cryo-EM structures; A.D. and D.L. generated the plasmids and purified the proteins used in this study; A.D. drafted the manuscript with input from W.H., D.J.T., D.L., C.M.O., and G.N. All authors reviewed the manuscript.

DECLARATION OF INTERESTS

D.J.T. and G.N. have equity interests in RAPPTA therapeutics. The remaining authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
PP2A - Aβ	Aviva Systems Biology	Cat#OAAB18901; RRID: AB_11223509
PP2A C Subunit - Methyl L309 (monoclonal)	Egon Ogris	Clone 7C10
Goat anti-rabbit IRDye	LI-COR Biosciences	Cat#926-32211; RRID: AB_621843
Goat anti-mouse IRDye	LI-COR Biosciences	Cat#926-68070; RRID: AB_10956588
Bacterial and virus strains		
BL21 (DE3)	New England Biolabs	Cat#C25271
DH10Bac	Thermo Fisher Scientific	Cat#10361012
DH5α	Thermo Fisher Scientific	Cat#18265017
Chemicals, peptides, and recombinant proteins		
TEV Protease (purified in-house)	Kapust et al.[45]	pRK793 (Addgene #8827)
DTT	Goldbio	Cat#27565-41-9
NHS	FortéBio	CAS#106627-54-7
EDC	FortéBio	CAS#25952-53-8
Protease Inhibitor Cocktail	Roche	Cat#05892791
SDS/PAGE 4–20% Polyacrylamide Gels	Bio-Rad	Cat#4561096
IPTG	Goldbio	Cat#I2481C
S-Adenosyl methionine (SAM)	Sigma-Aldrich	CAS#52248-03-0
DiFMUP	AAT Bioquest	Cat#11627
GST-PP2A-Aα	This paper	N/A
GST-PP2A-Aβ	This paper	N/A
GST-B56α	This paper	N/A
GST-B56γ	This paper	N/A
GST-LCMT1	This paper	N/A
Critical Commercial Assays		
Protein Thermal Shift Dye Kit	Applied Biosystems	Cat#4461146
Deposited data		
Cryo-EM map of Aα-B56γ-Cα	This paper	EMDB: 48300
Atomic coordinates of Aα-B56γ-Cα	This paper	PDB: 9MIP
Cryo-EM map of Aβ-B56γ-Cα	This paper	EMDB: 48224
Atomic coordinates of Aβ-B56γ-Cα	This paper	PDB: 9MF5
Oligonucleotides		
Aβ N-terminal deletion mutagenesis: F-GCAGCGGGTGGAGATGGAGATGATT	This paper	N/A
Aβ N-terminal deletion mutagenesis: R-GCCCTGAAAATACAGGTTTTCGGATCCACG	This paper	N/A
Aβ N-terminal addition to Aα (AαChimera) mutagenesis: F-CTCGGGACCGGCCCAGGAATGGCGG CGGCCGACGGC	This paper	N/A
Aβ N-terminal addition to Aα (AαChimera) mutagenesis: R-CTCTGATGCGCCCGCCATGCCC TGAAAATACAGGTTTTCGG	This paper	N/A
Recombinant DNA		
pGEX - Aα	This paper	N/A
pGEX - Aβ	This paper	N/A

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
pGEX - B56α	This paper	N/A
pGEX - B56γ	This paper	N/A
pGEX - LCMT1	This paper	N/A
Software and algorithms		
cryoSPARC v3.3.1	Punjani et al.[46]	https://cryosparc.com/
Serial EM	Mastronarde[47]	https://bio3d.colorado.edu/SerialEM/
Phenix v1.20	Liebschner et al.[48]	https://phenix-online.org/
COOT v9.6.0	Emsley et al.[49]	https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/
ChimeraX	Pettersen et al.[50]	https://www.cgl.ucsf.edu/chimerax/
EMReady	He et al.[51]	http://huanglab.phys.hust.edu.cn/EMReady/
ISOLDE	Croll[52]	https://tristanic.github.io/isolde/
NAMD v3.0.1	Phillips et al.[53]	https://www.ks.uiuc.edu/Research/namd/
AmberTools23	Case et al.[54]	https://ambermd.org/index.php
AlphaFold v3.0	DeepMind	https://github.com/google-deepmind/alphafold3
GROMACS v2024.3	Abraham et al.[55]	https://www.gromacs.org/
RELION v3.1	Zivanov et al.[56]	https://github.com/3dem/relion?tab=readme-ov-file
TOPAZ	Bepler et al.[57]	https://cb.csail.mit.edu/topaz/
GraphPad Prism v10	Graphpad	https://graphpad.com/
Octet Analysis Studio v12.2	Sartorius	https://www.sartorius.com/en/products/biolayer-interferometry/octet-r8
StepOne	Applied Biosystems	https://www.thermofisher.com/us/en/home/life-science/pcr/real-time-pcr/real-time-pcr-instruments/step-one-real-time-pcr-systems.html
LabSolutions	Shimadzu	https://www.shimadzu.com/an/products/software-informatics/labsolutions-series/index.html
ImageJ	National Institutes of Health	https://imagej.net/ij/
MolProbity	Chen et al.[58]	https://www.phenix-online.org/documentation/reference/molprobity_tool.html
VMD	Humphrey et al.[59]	https://www.ks.uiuc.edu/Research/vmd/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell culture

The *E. Coli* strain DH5α was used for all molecular cloning. The *E. coli* expression strain BL21(DE3) was used for the *in vitro* production of the PP2A Aα, Aβ, B56α, and B56γ subunit genes. The *E. coli* strain DH10Bac was used for the generation of an expression bacmid containing the human PP2A Cα subunit gene for subsequent transfection and baculovirus generation in insect cells. The Sf9 insect cell line was used for baculovirus generation and propagation following transfection with purified bacmid DNA. The High Five (*Trichoplusia ni*) insect cell line was used for expression of the human PP2A Cα subunit following infection with baculovirus.

METHOD DETAILS

Molecular dynamics simulations

All-atom molecular dynamics simulations were performed starting from initial structures using *de novo* prediction in AlphaFold v3.0 (DeepMind)[60] or those modeled into our cryo-EM structures and with the disordered Aβ N-terminus added using AlphaFold. The AMBERff19SB force field was used to describe the system with explicit OPC water molecules.[61,62] The system was placed in a rectangular water box with 16 Å padding and 25 charge-neutralizing sodium ions (23 for Aα) with additional salt (Na^+, Cl^-) around the

solute, resulting in a 150 mM salt condition, using the Amber23 LEaP module.[54] The water box padding size was confirmed by measuring the distance from the protein boundary to the water box edge in each direction. The distances are in the range of 18–19.5 Å, slightly exceeding the anticipated 16 Å padding. After initial system preparation, the MD simulations were performed using NAMD.[53] The systems were energy minimized for 10,000 steps to remove bad contacts, and then the systems were equilibrated with all heavy atoms restrained harmonically. During equilibration, the temperature was raised 10 K per 10,000 steps starting from 0 to 300 K using temperature reassignment. After reaching the desired temperature, harmonic restraints were gradually reduced using a scale from 1.0 to 0 with a 0.2 decrement for every 50,000 steps. MD simulations were performed under the NPT ensemble.[63,64] Langevin dynamics was used for constant temperature control, with the value of the Langevin coupling coefficient and the Langevin temperature set to 5 ps and 300 K, respectively. The pressure was maintained at 1 atm using the Langevin piston method with a period of 100 fs and decay times of 50 fs. All RMSD analyses were performed using the RMSD trajectory tool in VMD software, and RMSF analyses were performed using GROMACS software version 2023.3.[55,59] The reference frame for all analyses was the starting structure (frame 1). The resulting data was then plotted using GraphPad Prism 10. Smoothed trajectory movies were recorded using VMD. Trajectory smoothing was performed using VMD and a smoothing window of 10.[59]

Protein expression, purification, and complex assembly

All constructs were generated using Gibson assembly, and chimeras were obtained using a standard PCR-based cloning strategy.[65] Aα, Aβ, B56α, and B56γ were overexpressed using the BL21(DE3) bacterial strain with a glutathione S transferase (GST)-fusion tag at the N-terminus followed by a TEV site. Protein expression was induced by adding 0.1 mM IPTG at OD 0.6, and cells were grown for an additional 18 h at 18°C. Cells were harvested by centrifugation, resuspended in purification buffer (50 mM HEPES (pH 8.0), 250 mM NaCl, and 2 mM DTT), then lysed with a French press, and spun at 36,000 × g for one hour. The soluble fraction was incubated with GST affinity beads pre-equilibrated with purification buffer, at 4°C for one hour. After incubation, the beads were extensively washed with purification buffer on a gravity filter column and then resuspended to 5 mL in purification buffer. Dialysis and cleavage were performed overnight through the addition of 80 μg of TEV and transfer of the sample to a dialysis bag that was placed in a 5 L solution of 50 mM HEPES (pH 8.0), 50 mM NaCl, and 1 mM DTT at 4°C. The elution fraction was collected using a gravity filter column and then applied to a HiTrap Q HP 5 mL anion exchange chromatography column pre-equilibrated with 50 mM HEPES (pH 8.0), 50 mM NaCl, and 1 mM DTT for further purification. The Cα gene was cloned into pFastBac1 with an N-terminal 6xHis tag, expressed in Hi5 (*Trichoplusia ni*) insect suspension cells using recombinant baculovirus infection, affinity purified using Ni-NTA beads and further purified by IEX and SEC. The buffer used for purification was 50 mM HEPES (pH 8.0), 100 mM NaCl, 3 mM β-Mercaptoethanol, and 4 mM imidazole. The Aα and AβB56γC holoenzyme complexes were assembled by mixing highly purified A, C, and B56γ at a 1:1.1:1.5 ratio and subsequently purified using a Superdex 200 SEC column equilibrated with HBS buffer (20 mM HEPES, 150 mM NaCl, and 1 mM DTT, pH 7.5).

Analytical size-exclusive chromatography

The Aα and Aβ holoenzyme and core dimer complexes were assembled and purified as described and diluted to the desired concentration in HBS buffer (20 mM HEPES, and 150 mM NaCl, and 1 mM DTT, pH 7.5). All SEC experiments were performed using an HPLC system (Shimadzu) equipped with a multi-wavelength UV detector, and the indicated column was equilibrated using HBS. For the analytical SEC experiments, 10 μL samples from each of the unconcentrated primary Aβ IEX peak fractions were loaded on a Superdex 200 increase PC3.2 analytical column (GE Healthcare). For the C-subunit binding experiment of Aβ IEX peak 2, the concentration of the IEX peak fraction was measured by nanodrop, and then 10-fold molar excess C-subunit (12 μL) was added to a 2 μL sample of the IEX peak fraction, before loading on a Superdex 200 increase PC3.2 analytical column (GE Healthcare). For the aggregation over time analytical SEC experiments, a 100 μL 1 μM stock was prepared using HBS from each of the purified and concentrated aliquots of Aα, Aβ, AαChimera, and AβNT. These stocks were placed in a 37°C water bath, and a 15μL sample was taken every hour for 6 h. A 15 μL sample from each stock was taken before incubation in the water bath for the T_0 time point. Due to each SEC run taking 80 min, the samples were placed in the Shimadzu autosampler block chamber set to maintain a temperature of 4°C while in the queue, after which they were loaded on a Superdex 200 increase PC3.2 analytical column (GE Healthcare). SEC peak positions, heights, and peak integrations were determined using LabSolutions software (Shimadzu).

Protein thermal shift assay

Thermal denaturation assays were performed using Applied Biosystems Protein Thermal Shift dye, and thermal denaturation was monitored by differential scanning fluorimetry. Briefly, the protein was diluted to a final assay concentration of 1 μM in HBS buffer pH 7.5, containing 1:1000× dye, with a final assay volume of 20 μL. Heat denaturation curves were recorded using a real-time PCR instrument (Applied Biosystems, StepOne), applying a temperature gradient of 2°C/min. Analysis of the data was performed using default StepOne Software. Melting temperatures were determined by the derivative method using the inflection points of the plots of -d(RFU)/dT.

Biolayer interferometry

BLI measurements were performed on an Octet R8 Protein Analysis System (Sartorius). AR2G biosensors were hydrated in HBS (20 mM HEPES, 150 mM NaCl, and 1 mM DTT, pH 7.5) for 15 min and then automatically attached to the light sensors

of the instrument. Six biosensors were then simultaneously activated by incubation for 5 min with 75 mg/mL 1-ethyl-3-(3-dimethylaminopropyl)-carbodiimide hydrochloride (EDC) and 11.5 mg/mL N-hydroxysuccinimide (NHS) resuspended in water. Following activation, the biosensors were incubated with a purified PP2A B56 subunit (B56α diluted to 5.5 mg/mL in 50mM HEPES, 350 mM NaCl, 2 mM DTT, pH 7.5 or B56γ diluted to 1.5 mg/mL in 50 mM HEPES, 50 mM NaCL, and 1 mM DTT, pH 7.5) for 10 min. After immobilization, the biosensors were dipped into a 1 M Tris solution for 5 min to quench the amine coupling reaction, followed by incubation in HBS for 5 min to establish a baseline. The analytes (AαC, AβC, AαChimera-C, and AβNT-C dimers) were purified by SEC with Superdex 200GL 10/300 in HBS buffer. The PP2A AC dimers were prepared at a concentration range of 0–500 nM by 1:1 serial dilution in HBS buffer, with the highest concentration at 500 nM and lowest at 31.25 nM. Association was performed simultaneously by incubating the six biosensors with immobilized B56α or B56γ in six wells containing binding buffer control (HBS) and increasing PP2A A-C concentrations for 300 s, followed by a 600 s dissociation step performed in HBS buffer. After each dissociation measurement, the surface underwent three rounds of regeneration to compete off any remaining bound AC dimer by incubating the biosensor first in 5 M NaCl for 10 s, followed by neutralization in HBS buffer for 10 s. Data processing and analysis were completed using the Sartorius Octet BLI systems software. Data processing includes the subtraction of the binding buffer control channel. The binding affinities were calculated based on the on- and off rates by fitting the 1:1 binding model. Results for binding affinities are presented as averages with standard deviation determined from two independent experiments (n = 2, shown as error bars in Figure 5A and error values in 5B).

Cryo-EM data acquisition and image processing

Both the AαB56γC and AβB56γC complexes were freshly prepared by purifying the *in vitro* assembled complex using size-exclusion chromatography. 3.5 μL of the complex peak fraction was applied to a Quantifoil AU 1.2/1.3 300 mesh grid that was glow-discharged and coated with graphene oxide support film prepared in-house following a previously described protocol.[66] The grids were blotted for 4 s with a blot force −5 at 4°C and 100% humidity and plunge-frozen in liquid ethane using Vitrobot Mark IV (ThermoFisher). Movie stacks were recorded using a FEI Titan Krios transmission electron microscope G3i operated at 300 keV and equipped with a Gatan K3 direct electron detector and Gatan BioQuantum image filter operated in zero-loss mode with a slit width of 20 eV. Automated data collection was carried out using serialEM at a nominal magnification of 130,000× with a physical pixel size of 1.058 Å/pixel (0.529 Å/pixel at super-resolution) for the Aβ complex and 130,000× with a physical pixel size of 1.056 Å/pixel (0.528 Å/pixel at super-resolution) for the Aα complex. Movies were collected in 40 frames with a total electron dose of 40 e/Å^2. The defocus range was set in the range of −1 to −3 μM. All data processing steps were performed usv3.3.1. Micrographs were filtered to remove outliers in motion correction and/or CTF estimation results and screened to remove micrographs with significant non-vitreous ice contamination. For both datasets, the particles were automatically picked using the blob picker with a diameter of 130 Å. Reference-free 2D classification of the particles was performed with 200 classes and limited maximum resolution to 16 Å. Particles from the good 2D class averages were subjected to another round of 2D classification with 200 classes, and particles from classes with clear secondary structure features were selected. This particle set was then further refined in 3D by cycles of heterogeneous refinement and non-uniform refinement. The initial heterogeneous refinement was carried out using an ab initio volume generated from the selected particle set and three ab-initio volume "decoy" classes generated from obvious "junk" classes resulting from the first round of 2D classification. For the Aβ dataset, the initial map showed poor resolution of B56γ. Therefore, the particles were used in 3D classification without a volume set to ten classes using PCA mode, the filter resolution set to 6 Å, and the parameter "number of components" set to 3. Two 3D Classes with improved resolution for B56γ were chosen. The two maps were used in another 3D classification job using all of the input particles for the previous 3D classification job. This 3D classification job used the same initialization settings as previously with the number of classes set to two and the filter resolution set to 4 Å. The map with superior B56γ resolution was chosen and refined using non-uniform and local refinement. Micrographs containing particles from this map spanning a wide defocus range were manually selected and used to train a TOPAZ deep learning picking model.[57] This model was then used to repick particles from all micrographs. The particles were refined by cycles of 3D heterogeneous refinement and non-uniform refinement using the previously refined map obtained from 3D classification and the previously generated decoy volumes as the initial input. Map resolutions were determined by "gold standard" half-maps Fourier shell correlation (FSC) that was calculated and corrected for masking effects by cryoSPARC. The final map was sharpened using EMReady.[51] Local resolution variations were estimated using RELION-3.1.[56]

Model building and refinement

The AlphaFold predicted model of Aβ was aligned to Aα within the previously determined AαB56γC crystal structure (PDB ID: 3FGA).[36] Aα was then removed and the resulting assembled PDB containing Aβ was used as the starting point for refinement. This model was additionally used as the starting point for AαB56γC refinement. The models were built and refined into the respective maps by iterating between manual rebuilding and refinement in Coot (version 9.6.0) and ISOLDE, and automated global real-space refinement in Phenix.[48,49,52] Map validations were performed using PHENIX, and the final atomic model was evaluated using MolProbity.[58] Figure panels depicting cryo-EM maps or atomic models were generated using ChimeraX.[50]

Negative-stain electron microscopy

All samples were added to 200 mesh copper grids with a formvar carbon film (EMS CF200-CU) and stained using 2% uranyl formate. Each grid was made using a sample from the main elution fraction from both peak 1 and peak 2 of the Aβ ion-exchange

chromatography chromatogram. Micrographs were acquired on a Tecnai TF20 electron microscope (FEI Co.) operating at 200 keV. The TF20 was equipped with TVIPS F-416 CMOS (4k ×4k) camera, and images were acquired at a magnification of 50,000×.

QUANTIFICATION AND STATISTICAL ANALYSIS

The details of the quantification and statistical analyses are included in figure legends or the relevant sections of the method details. Cryo-EM data collection and refinement statistics are summarized in Table S1.

Structure, Volume *33*

Supplemental Information

Regulatory mechanisms of PP2A complex assembly driven by physicochemical differences in A-subunit isoforms

Alexander Day, Wei Huang, Daniel Leonard, Caitlin M. O'Connor, Goutham Narla, and Derek J. Taylor

Supplemental Information

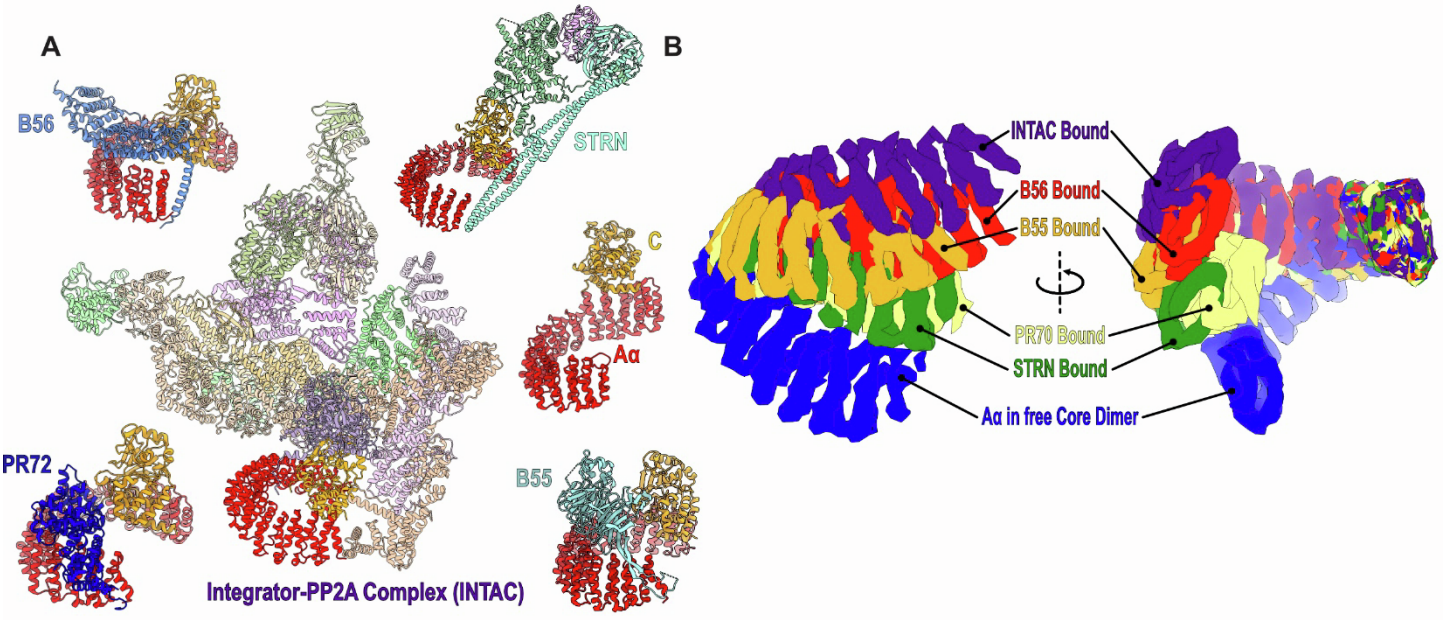

Figure S1. PP2A A-subunit conformational dynamics as a monomer and bound in complex, related to Figure 1. (A) Existing PP2A holoenzyme models determined by X-ray crystallography and cryo-EM. PP2A core dimer (PDB: 2IE4). PP2A:B55 (PDB: 3FGA). PP2A:B56 (PDB: 3DW8). PP2A:PR70 (PDB: 4I5L). PP2A:STRN (PDB: 7K36). PP2A:INTAC (PDB:7CUN). **(B)** Superimposed Aα subunit models aligned to the constant A-C interface. Each A-subunit conformation is observed as part of a holoenzyme complex with a different B-subunit family. A-subunits are colored differentially depending on the bound PP2A B-subunit family member.

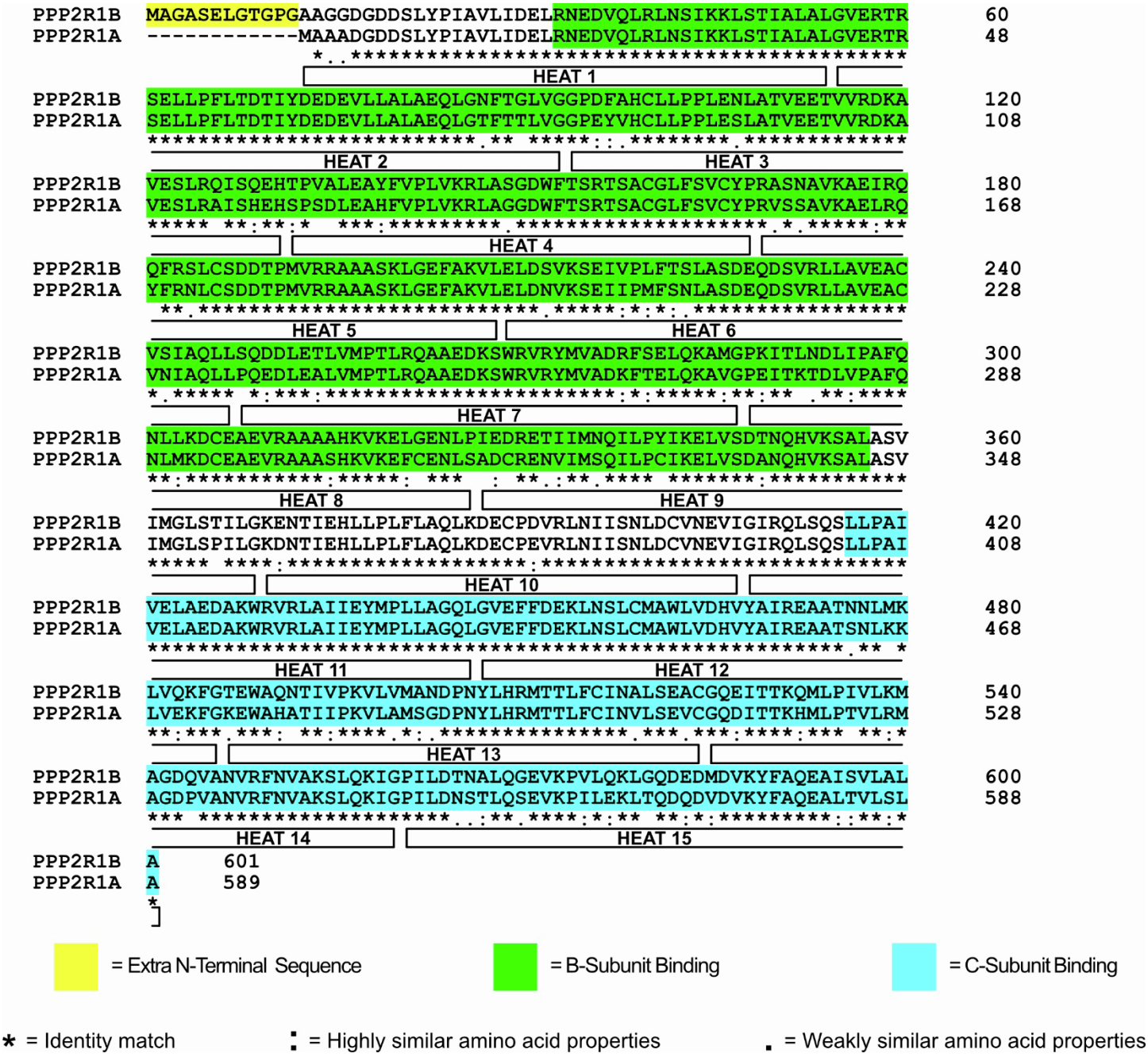

Figure S2. Sequence alignment of PP2A A-Subunit Isoforms, related to Figure 1 and Figure 3. Sequence alignment of the canonical amino acid sequences of the Aα and Aβ isoforms. Regions colored for extra N-terminal sequence in Aβ (yellow), B-subunit binding heat repeats (2-7) (green), and C-subunit binding heat repeats (11-15) (cyan). HEAT repeat regions are identified with boxes below the sequence with HEAT repeat number.

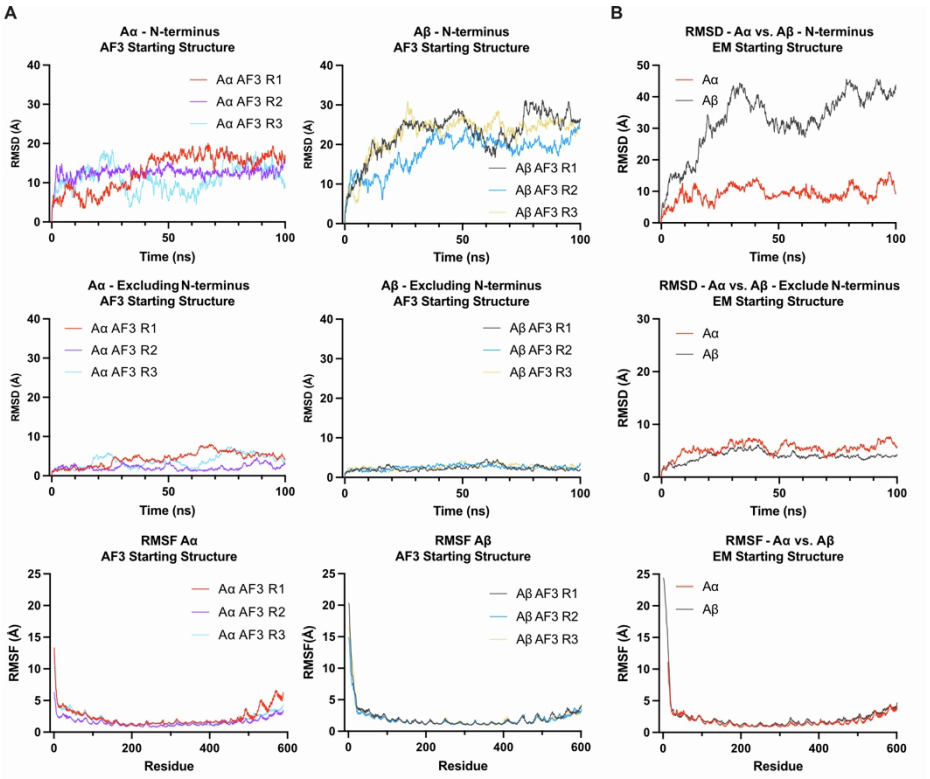

Figure S3. Molecular Dynamics Simulations, related to Figure 1. Three replicate molecular dynamics simulations were performed starting from the AF3 predicted structure of Aα or Aβ and simulated for 100ns. **(A)** RMSD plots of the N-terminal region (1-13 for Aα and 1-25 for Aβ) and excluding the N-terminus (14-589 for Aα and 26-601 for Aβ included) are shown for each replicate. RMSF plots are shown per residue for all replicates. VMD was used to perform RMSD calculations and GROMACS was used to perform RMSF calculations. **(B)** A single molecular dynamics simulation starting from our Aα and Aβ cryo-EM determined models with AF3 predicted N-terminal regions for both proteins. Simulations were run with the same parameters as those used in panel A. RMSD and RMSF plots were determined as described above and shown comparing the two simulations.

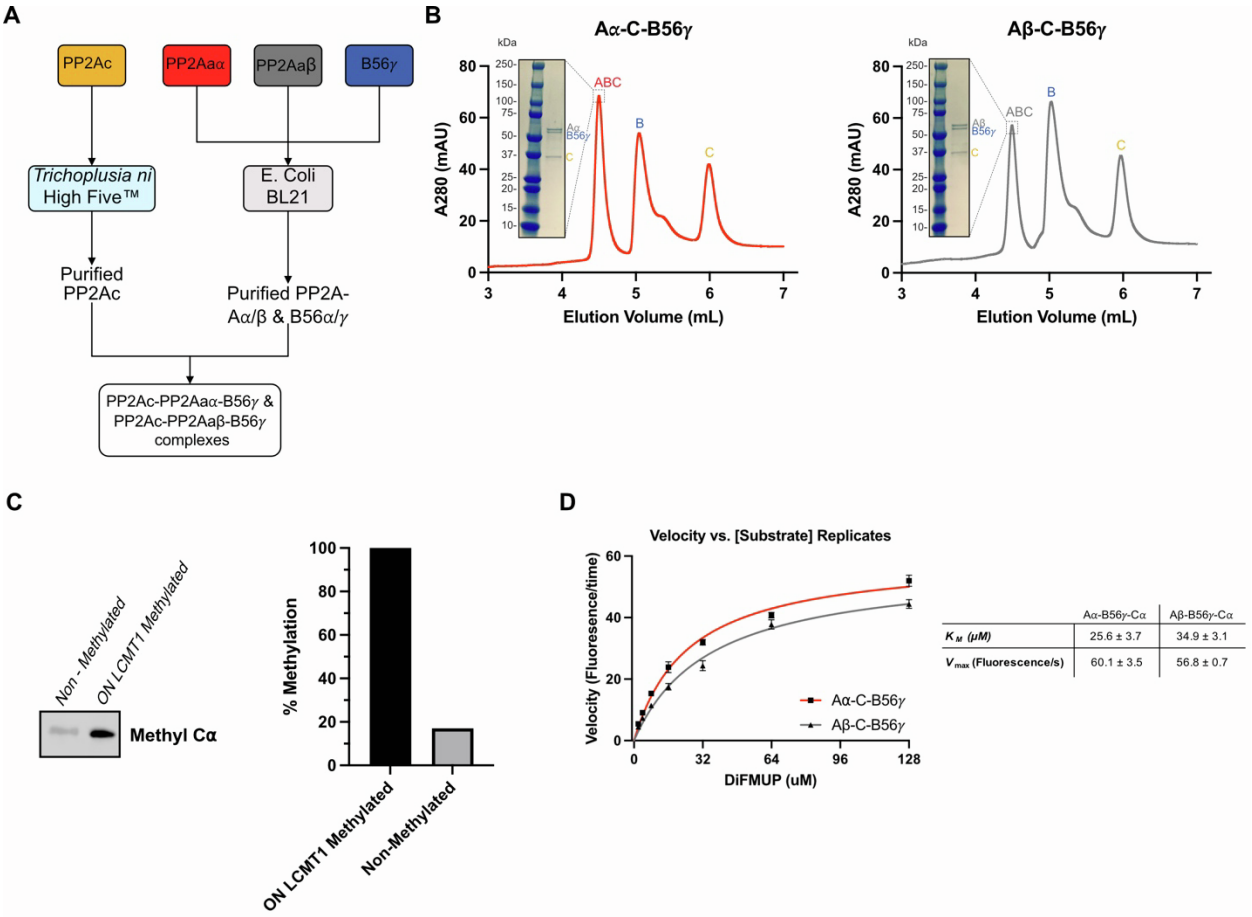

Figure S4. PP2A Complex Assembly, Purification, Activity and Methylation Quantification, related to Figure 2, Figure 3, and Figure 4. (A) Protein expression and purification scheme describing the expression system used to produce recombinant proteins. **(B)** Size exclusion chromatography (SEC) curves from in vitro assembly and purification of the PP2A:Aα-B56γ and PP2A:Aβ-B56γ holoenzymes. Coomassie-stained gels are shown for the SEC fraction collected from the trimer peak isolated from the SEC step. **(C)** Quantification of methylation modification of purified PP2A C-subunit as determined by western blot using an antibody (7C10; a gift from Dr. Egon Ogris) specific for the methylated C-subunit. Methylation quantification included the overnight incubation with purified, recombinant LCMT and excess S-Adenosyl methionine as the methyl donor. This sample was used as a "100% methylated C" and was compared to the non-methylated sample for quantification. It is approximated that ~17% of the human C subunit purified from insect cells is methylated at the terminal Leucine 309 residue. **(D)** DiFMUP phosphatase activity assay. Reactions were performed at a final AC core dimer concentration of 20 nM with excess (200 nM) B56γ. Increasing concentrations of the DiFMUP substrate (2-128 μM) were used in the reactions, which were performed in triplicate. A linear fit of each reaction curve provided the reaction velocity, which was plotted against DiFMUP concentration and fit with a non-linear Michaelis-Menten equation to determine the K_m and V_{max} of each complex for the DiFMUP substrate. Plot is the mean velocity from three replicates at each concentration point, with the error bars representing SD. Chart shows the mean values for the K_m and V_{max} calculated from individual non-linear curve fits from each of three replicates with the error values reported as SD. All calculations and analyses were performed using GraphPad Prism 10.

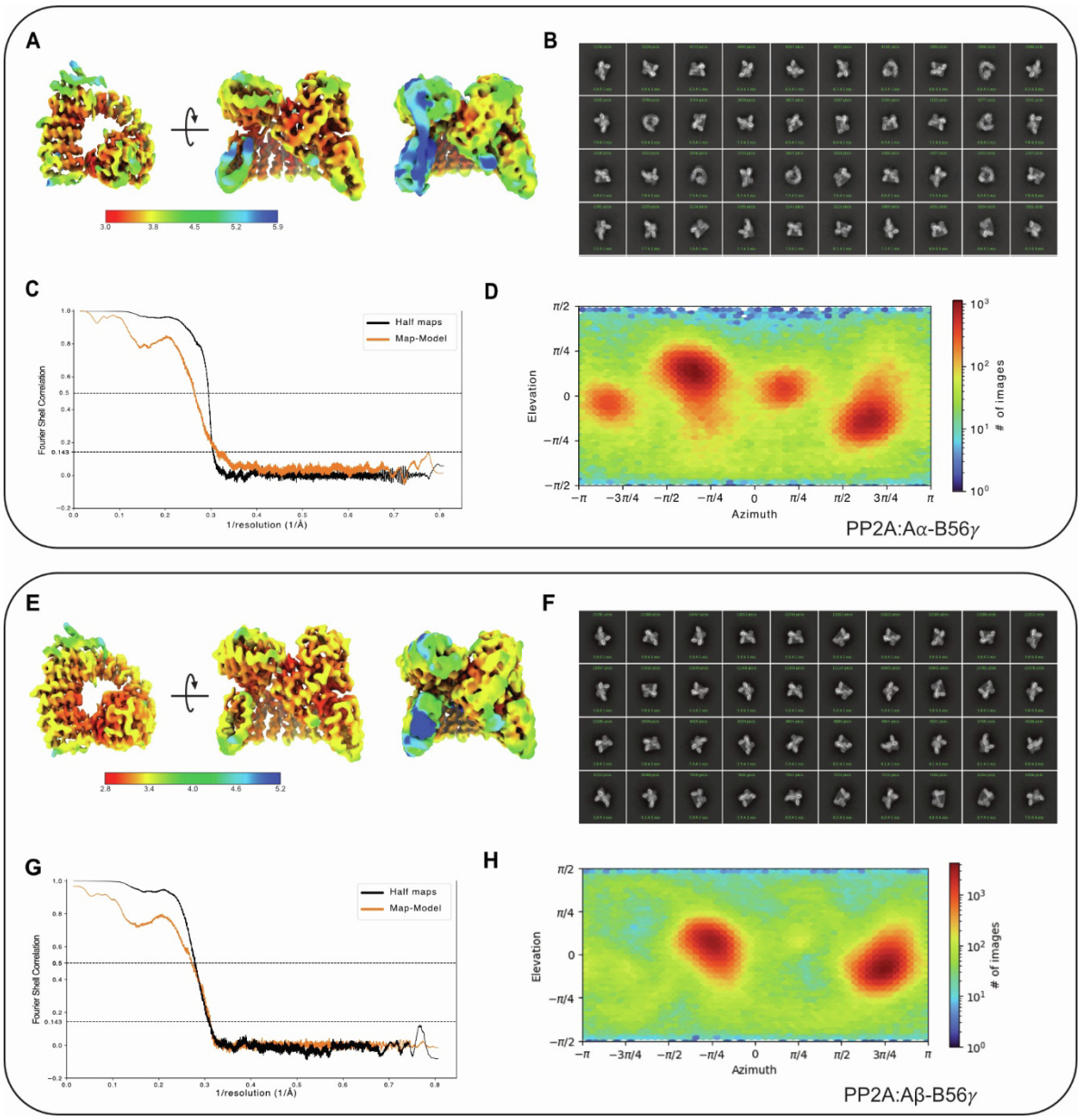

Figure S5. Image Processing of PP2A:Aβ-B56γ and PP2A:Aα-B56γ cryo-EM datasets, related to Figure 2, Figure 3, and Figure 4. (A) PP2A:Aα-B56γ refined map colored by local resolution. **(B)** Representative 2D classes from the final particle set comprising the refined PP2A:Aα-B56γ map. **(C)** The Fourier shell correlation (FSC) for the independently refined half-maps (black) and the full map and atomic model (orange) from the PP2A:Aα-B56γ dataset. The "gold standard" half-maps FSC was calculated and corrected for masking effects using cryoSPARC. The map-model FSC was calculated by PHENIX using a mask around the model based on map resolution. The FSC = 0.5 and 0.143 thresholds are marked by dashed lines. **(D)** Viewing direction distribution for the final PP2A:Aα-B56γ refined map generated by cryoSPARC local refinement. **(E)** PP2A:Aβ-B56γ refined map colored by local resolution. **(F)** Representative 2D classes from the final particle set composing the refined PP2A:Aβ-B56γ map. **(G)** The Fourier shell correlation (FSC) for the independently refined half-maps (black) and the full map and atomic model (orange) from the PP2A:Aβ-B56γ dataset. The "gold standard" half-maps FSC was calculated and corrected for masking effects using cryoSPARC. The map-model FSC was calculated by PHENIX using a mask around the model based on map resolution. The FSC = 0.5 and 0.143 thresholds are marked by dashed lines. **(H)** Viewing direction distribution for the final PP2A:Aβ-B56γ refined map generated by cryoSPARC local refinement.

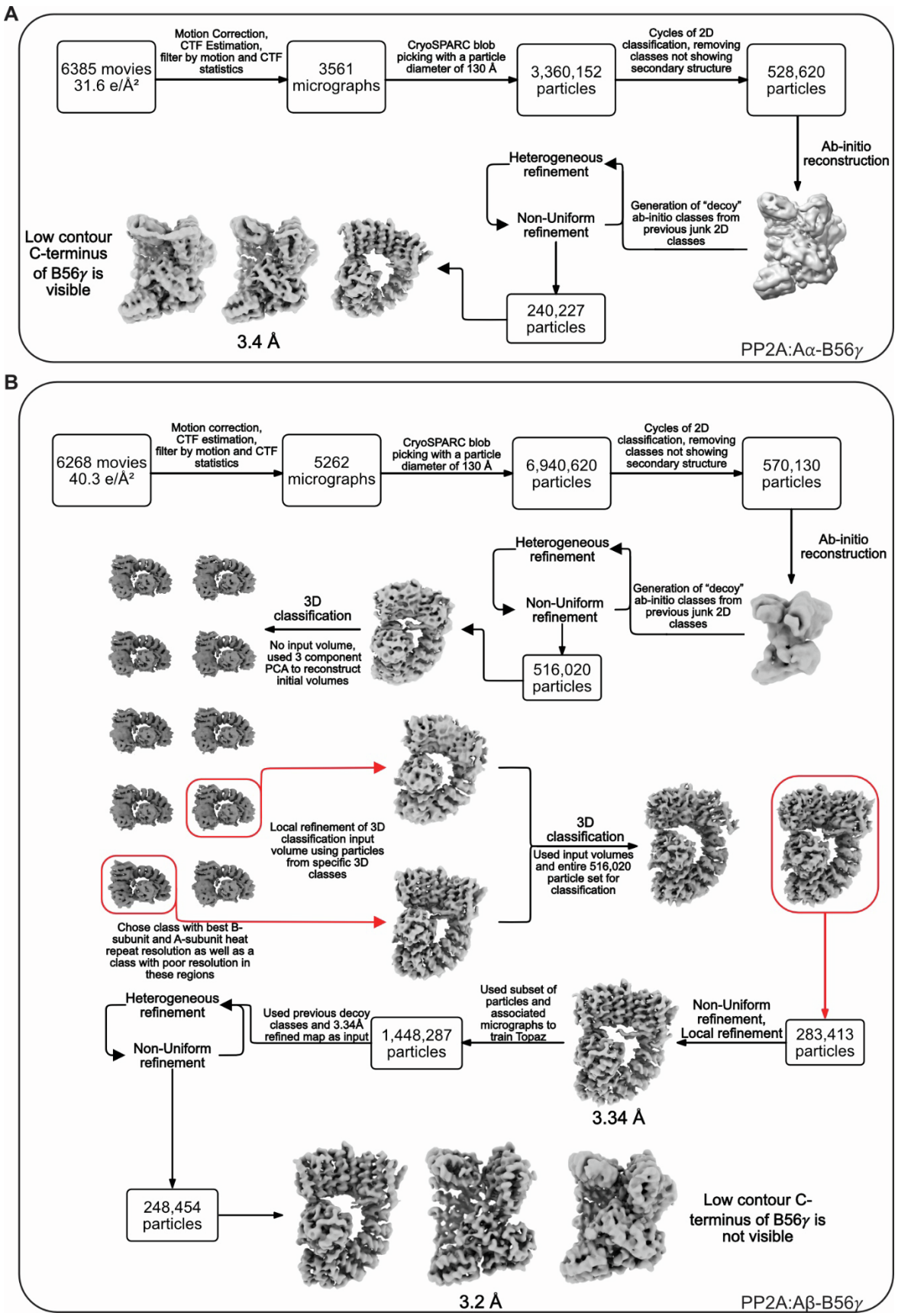

Figure S6. Cryo-EM image processing workflow for PP2A:Aα-B56γ and PP2A:Aβ-B56γ, related to Figure 2, Figure 3, and Figure 4. Cryo-EM and sorting procedures are described for **(A)** PP2A:Aα-B56γ and **(B)** PP2A:Aβ-B56γ complexes. Particle counts and reconstruction resolutions are given at key junctions. All resolutions noted are calculated by the gold standard half-maps FSC=0.143 criterion.

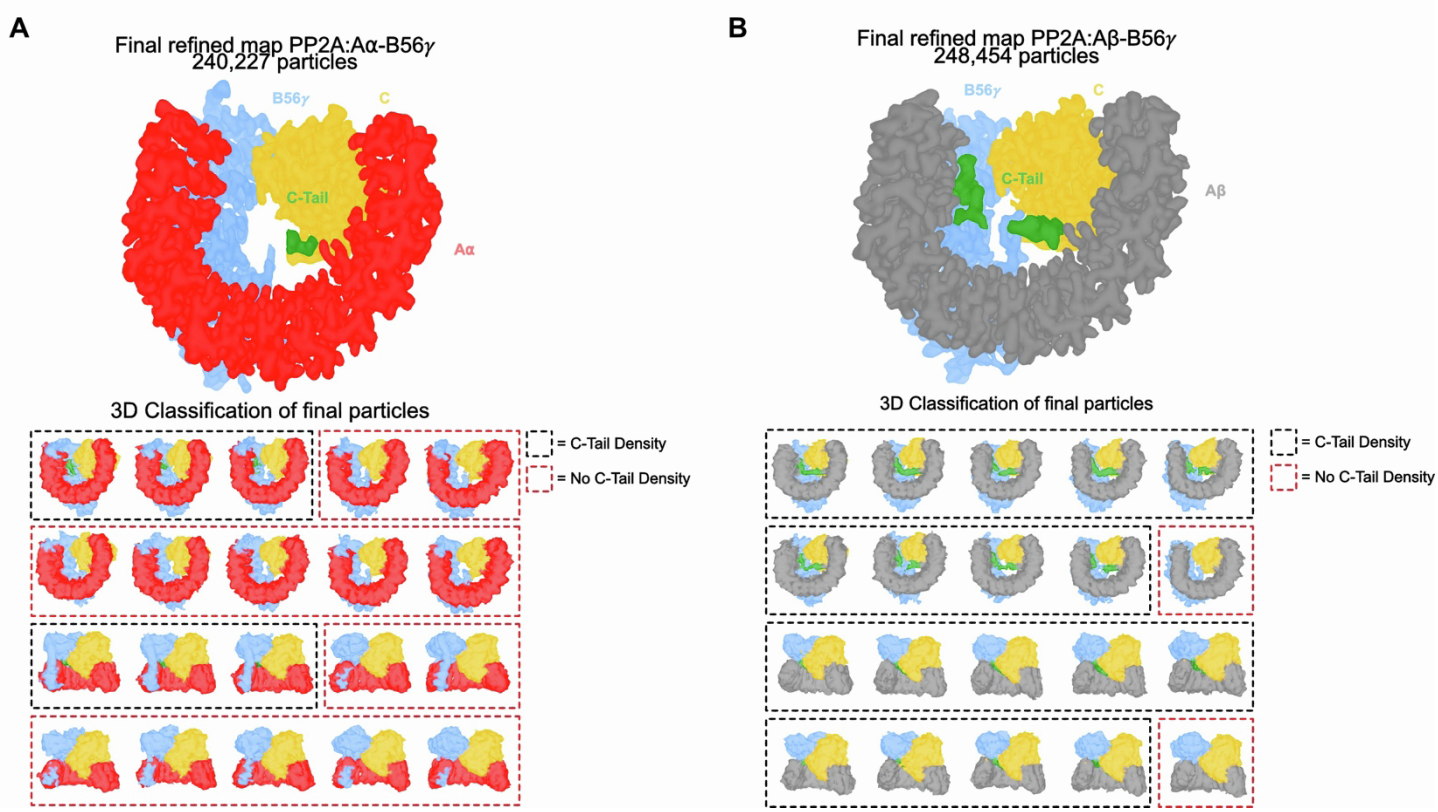

Figure S7. 3D classification of particles from final refined PP2A:Aα-B56γ and PP2A:Aβ-B56γ cryo-EM maps, related to Figure 4. 3D classification of particles from the final refined cryo-EM maps from both the **(A)** PP2A:Aα-B56γ and **(B)** PP2A:Aβ-B56γ datasets. 3D classification performed in cryoSPARC into 10 classes, with a filter resolution of 6 Å. Classes in which PP2A C-tail density is observed are outlined with a black dotted line, and those without, a red dotted line. PP2A C-tail density is colored green. Initial refined map and associated particles used for input into cryoSPARC 3D classification job shown at the top of each figure.

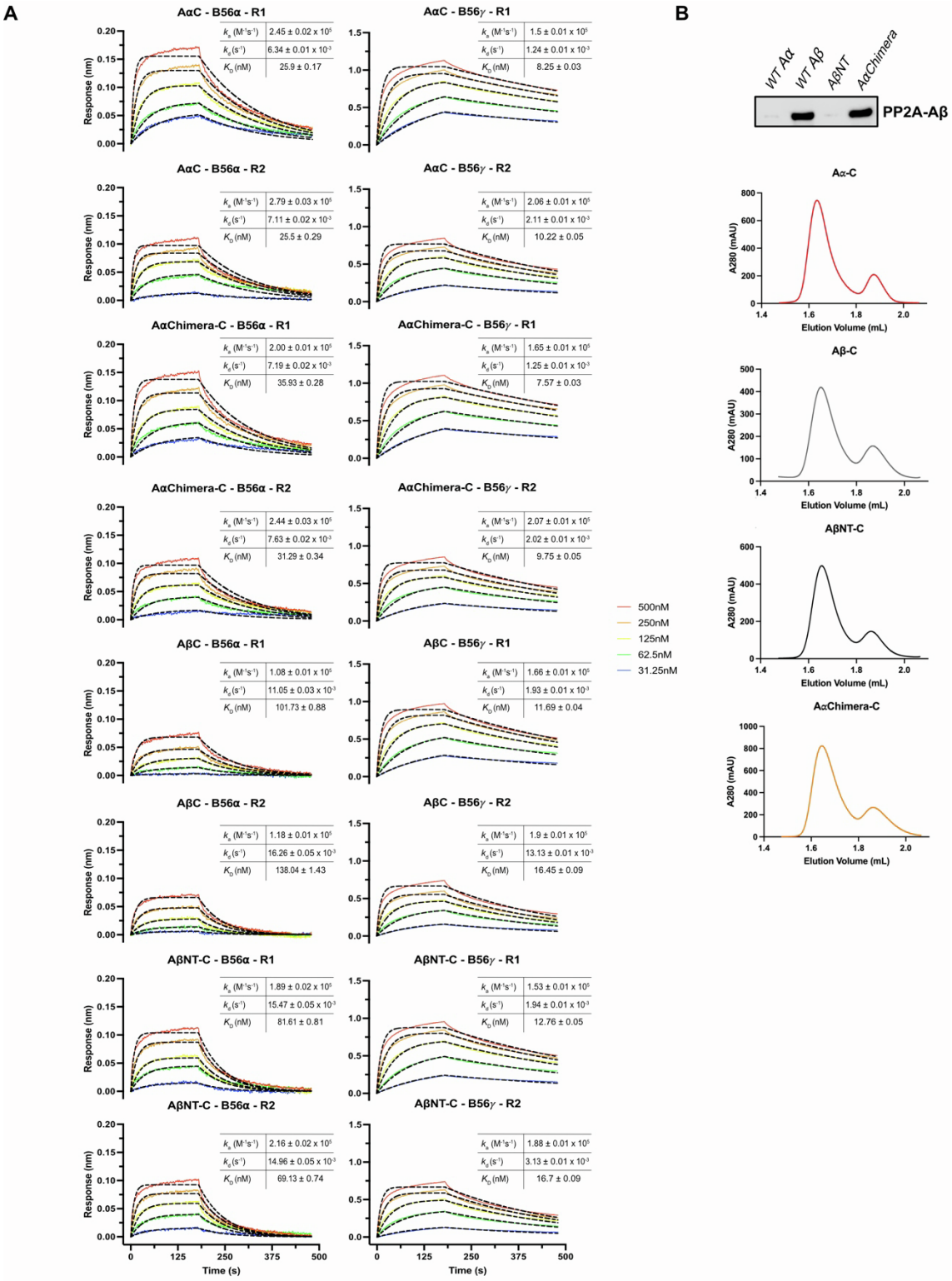

Figure S8. Biolayer interferometry (BLI) response plots from assays of Aα-C, Aβ-C, AαChimera-C, and AβNT-C binding to B56α and B56γ, related to Figure 5. (A) Raw (colored curves) and fit BLI sensograms (black dotted curve) for each of the two repeats for each core dimer and B56-isoform tested. Different colors represent different concentrations of the core dimer that were serially diluted 4x from red (500 nM) to blue (31.25 nM) to result in the five curves per plot. Values and error for the association rate constant (k_a), dissociation rate constant (k_d) and equilibrium dissociation constant (K_D) are reported for each experiment resulting from the fit of 5 sensograms using the 1:1 binding model with the Sartorius Octet BLI systems software. Reported errors are mean ± SEM for both the k_a and k_d values calculated directly from curve fitting. K_D error values are calculated from the k_a and k_d error using propagation of uncertainty. **(B)** Validation of the AβNT and AαChimera constructs using an antibody specific to the Aβ N-terminus (top). SEC curves from in-vitro assembly and purification of the Aα-C, Aβ-C, AαChimera-C, and AβNT-C core dimer complexes use for BLI experiments.

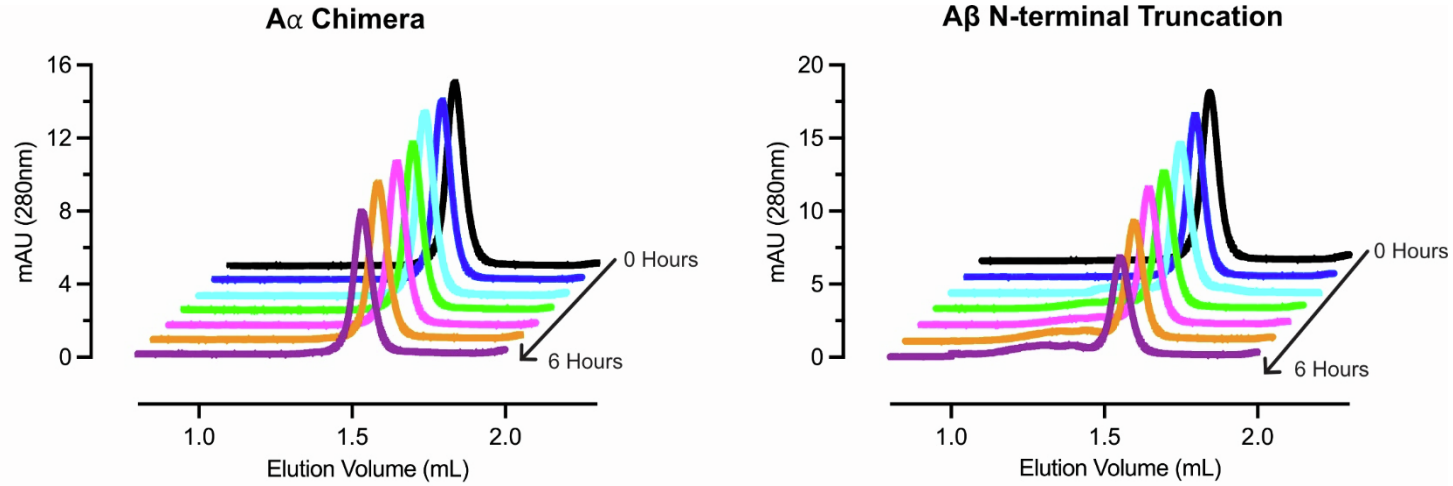

Figure S9. Aggregation over time of AβNT and AαChimera, related to Figure 7. Equal concentrations of AβNT and AαChimera were incubated at 37°C, and samples were taken every hour for 6 hours and analyzed by size exclusion chromatography (SEC). The resulting curves for AβNT and AαChimera are plotted. Different colors correspond to different time points.

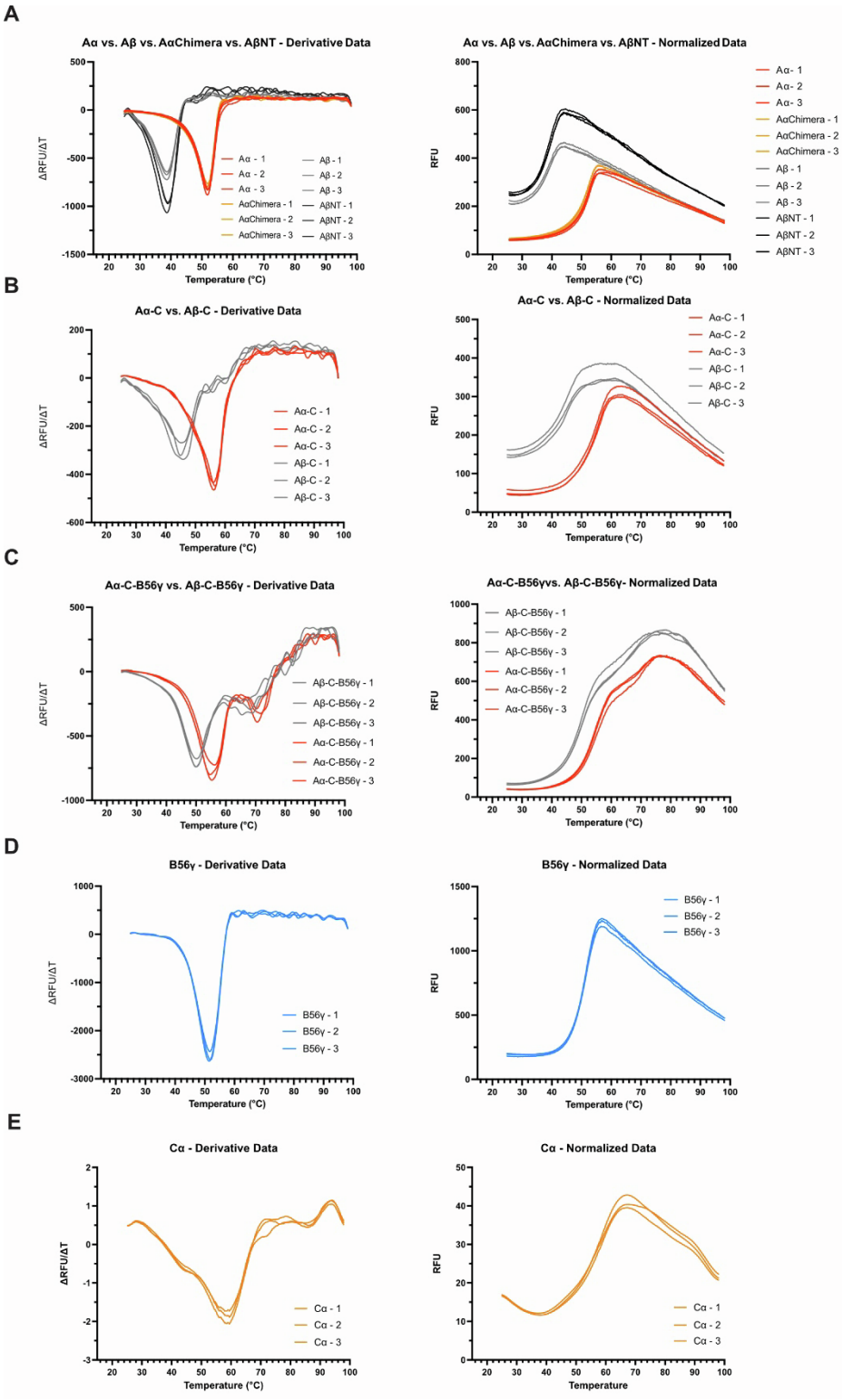

Figure S10. Thermal Shift Analysis normalized and derivative curves, related to Figure 7. Thermal shift analyses were performed on the A-subunit isoform monomers **(A),** core dimers **(B),** PP2A:B56γ holoenzyme trimers **(C)**, and B56γ monomer **(D)** as well as C subunit monomer **(E)** as controls. Each measurement was performed in triplicate. The inflection point of derivative plots was used to determine the melting point for each PP2A assembly state tested.

Table S1. Cryo-EM data collection, refinement and validation statistics, related to Figure 2, Figure 3, and Figure 4.

	Aα-B56γ-Cα (EMDB: EMD-48300) (PDB: 9MIP)	**Aβ-B56γ-Cα (EMDB: EMD-48224) (PDB: 9MF5)**
Data collection and processing		
Magnification	130,000x	130,000x
Voltage (kV)	300	300
Electron exposure (e-/Å)	31.59	40.32
Defocus range (μm)	-1 to -3	-1 to -3
Pixel size (Å)	1.056	1.058
Symmetry imposed	C1	C1
Initial particle images (no.)	3,360,152	6,940,620
Final particle images (no.)	240,227	248,454
Micrographs (no.)	3,561	5,262
Map resolution (Å) FSC threshold	3.4 FSC = 0.143	3.2 FSC = 0.143
Map resolution range (Å)	5.87 to 3.04	5.22 to 2.82
Refinement		
Initial model used (PDB)	3FGA	3FGA (AlphaFold for Aβ)
Model resolution (Å) FSC threshold	3.8 FSC = 0.5	3.6 FSC = 0.5
Model composition Non-hydrogen atoms Protein residues Ligands	 10549 1312 MN: 2	 9993 1244 MN: 2
B factors (Å^2) (min/max/mean) Protein Ligand	 121.42/408.29/193.58 188.16/221.76/204.96	 105.23/264.68/170.69 152.45/193.62/173.03
R.m.s deviations Bond lengths (Å) Bond angles (°)	 0.010 1.291	 0.009 1.241
Validation MolProbity score Clashscore Poor rotamers (%)	 1.25 4.88 0	 1.15 3.59 0
Ramachandran plot Favored (%) Allowed (%) Disallowed (%)	 98.08 1.92 0	 98.22 1.78 0